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Principles of Gamma Synchrony Predict Figure–Ground Perception in Texture Stimuli

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eLife Assessment

Karimian et al. present a **valuable** new model to explain how gamma-band synchrony (30–80 Hz) can support human visual feature binding by selectively grouping image elements, countering recent criticisms that the stimulus dependence of gamma oscillations limits their functional role. Grounded in the theory of weakly coupled oscillators the model captures behavioural patterns observed in human psychophysics, offering support for the potential role of synchrony-based mechanisms in feature-binding. The development of the model in alignment with primate electrophysiology **convincingly** supports the paper's claims that gamma synchrony may be the underlying mechanism. While the paper does not present electrophysiological results that directly link gamma oscillations to figure-ground segregation in the presented task, the model makes several predictions that can be tested experimentally.

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Abstract

Gamma synchrony is ubiquitous in visual cortex, but whether it contributes to perceptual grouping remains contentious based on observations that gamma frequency is not consistent across stimulus features and that gamma synchrony depends on distances between image elements. These stimulus dependencies have been argued to challenge the idea that the visual system groups image elements by synchronizing the neural assemblies that encode them. Here we argue instead that these dependencies may shape synchrony in perceptually meaningful ways. Indeed, according to the theory of weakly coupled oscillators (TWCO), synchrony-based grouping mechanisms require stimulus dependence. Synchronization among coupled oscillators depends on frequency dissimilarity and coupling strength, which in early visual cortex relate to local feature dissimilarity and physical distance, respectively. We manipulated these factors in a texture segregation experiment wherein human observers identified the orientation of a figure defined by reduced contrast heterogeneity compared to the background. Human performance followed TWCO predictions both qualitatively and quantitatively, as formalized in a computational model. Moreover, we found that when enriched with a Hebbian learning rule, our model also predicted human learning effects: Increases in model gamma synchrony due to perceptual learning predicted improvements in texture segregation across sessions. Taken together, our data suggest that the stimulus-dependence of gamma synchrony captures local image statistics and is linked to the stimulus-dependence of texture segregation, and that the effect of visual experience on gamma synchrony provides a viable perceptual learning mechanism for training-induced improvements

in texture segregation. Our results suggest that gamma synchrony with its inherent stimulus dependencies can provide a plausible mechanistic basis for perceptual grouping and visual scene segmentation.

Introduction

Oscillations are ubiquitous in the cortex (Buzsáki et al., 2013) and can synchronize both within and between cortical areas (Anand et al., 2023; Lowet, Roberts, Peter, Gips, & De Weerd, 2017; Melloni et al., 2007), but whether this contributes to neural information processing remains a matter of debate (Doelling & Florencia Assaneo, 2021; Duecker et al., 2021; Fernandez-Ruiz et al., 2023; Ray & Maunsell, 2015; Roelfsema, 2023). Early suggestions that synchrony in the gamma frequency band (30 – 80 Hz) plays a central role in visual feature binding (Singer, 1999; Uhlhaas et al., 2008) have been called into question based on observations that the gamma frequency is not consistent across stimulus features (Ray & Maunsell, 2010, 2015; Shirhatti et al., 2022) and depends on distances between image elements (Roelfsema, 2023; Roelfsema et al., 2004), making it difficult to group components of the same object by synchrony among the neural assemblies encoding these components (Dubey & Ray, 2020; Roelfsema, 2023). Alternatively, it has been proposed that the stimulus dependence of gamma synchrony facilitates, rather than hinders, their functional significance for visual processing by allowing contiguous neural assemblies that share a sufficiently similar oscillation frequency to synchronize into meaningful groups, while also blocking synchrony among assemblies with substantial frequency difference or physical separation (Lowet et al., 2015; Lowet, Roberts, Peter, Gips, & De Weerd, 2017). Here we show empirical and computational support for this view.

Analyzing a visual scene requires integration of features into coherent objects (feature binding), but also segregation of features belonging to distinct objects (feature separation). It remains unclear how this is achieved, but the stimulus dependence of gamma may be critical for a synchrony-based neural grouping mechanism that achieves both feature binding and separation. This idea is rooted in the theory of weakly coupled oscillators (TWCO), which describes the preconditions for synchrony among coupled oscillators (Acebrón et al., 2005; Ermentrout et al., 2019; Kuramoto, 1984; Neu, 1979; Strogatz, 2000). A group of coupled oscillators synchronizes if the discrepancy in their frequencies, referred to as their detuning, is overcome by the strength of their connection, referred to as their coupling. Thus, synchrony can occur even in the presence of strong detuning, if the coupling strength is sufficiently high, whereas if the coupling strength is low, synchrony can only occur if the detuning is also minimal. This relationship can be graphically depicted in an Arnold tongue (Coombes & Bressloff, 1999; Pikovsky et al., 2001), which shows the regions where synchrony occurs based on the balance between detuning and coupling strength (see Figure 1a for an illustration). These abstract principles are concretely realized in early visual cortex. Neural assemblies exhibit gamma oscillations in their population activity at frequencies that are directly related to stimulus features such as spatial frequency, contrast and orientation (Dubey & Ray, 2020; Henrie & Shapley, 2005; Shapira et al., 2017), and particularly contrast (Hadjipapas et al., 2015; Lowet et al., 2015; Roberts, Lowet, Brunet, TerWal, Tiesinga, Fries, & DeWeerd, 2013). In early visual cortical areas, coupling strength between neural assemblies is directly related to the efficacy of lateral anatomical connectivity, which declines with cortical distance (Boucsein et al., 2011; Gilbert & Wiesel, 1983; Lowet et al., 2015; Lowet, Roberts, Peter, Gips, & De Weerd, 2017; Stettler et al., 2002; Ts'o et al., 1986). In conjunction with the retinotopic organization of early visual cortex, this implies that neural assemblies encoding nearby visual regions are more strongly coupled. Taken together, synchrony in early visual cortex could occur across widely spaced neuronal assemblies in response to scenes with low feature heterogeneity, but only for closely spaced assemblies in response to scenes with high feature heterogeneity (see Figure 1b for an illustration). Indeed, a recent electrophysiological study in macaque V1 in which cortical distance and stimulus contrast heterogeneity were parametrically manipulated has confirmed that gamma synchrony behaves in line with the principles of TWCO (Lowet, Roberts, Peter, Gips, & De Weerd, 2017).

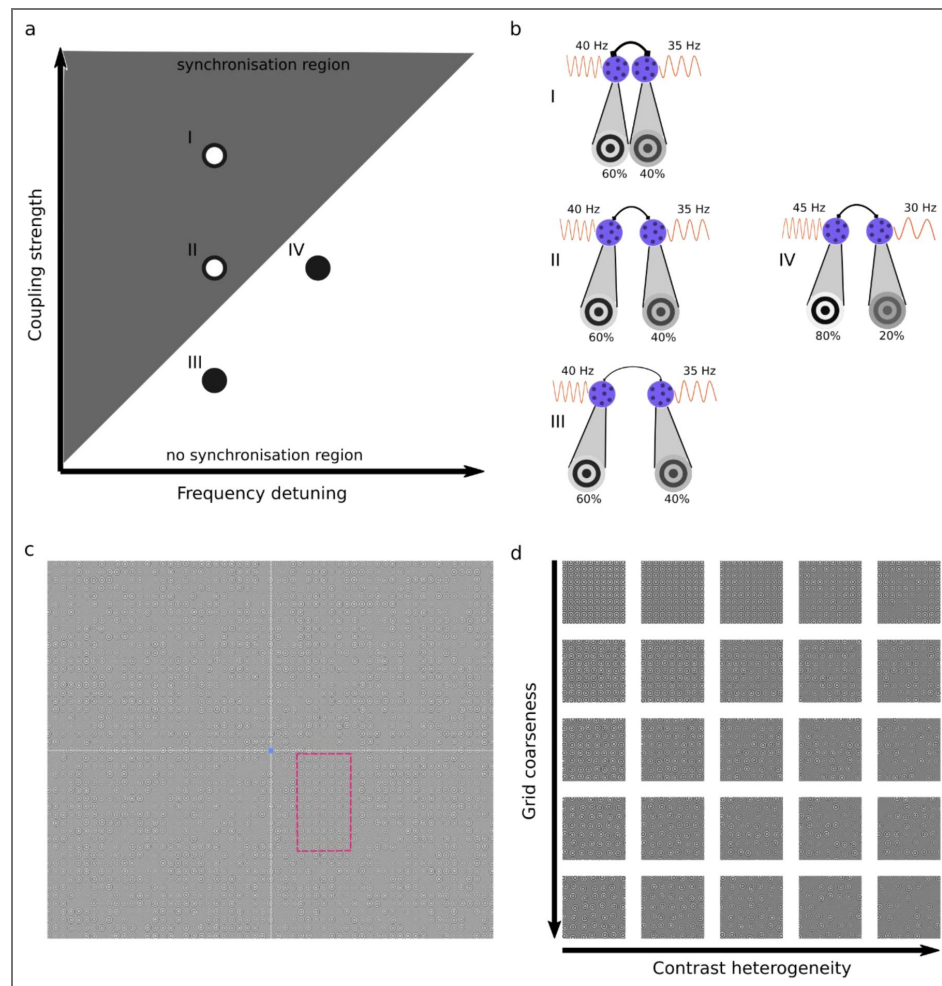


Figure 1. Schematic illustration of synchronization principles in visual cortex and stimulus design.

a, Arnold tongue: triangular region shows combinations of detuning and coupling strength that allow synchrony (light grey). Open circles indicate two scenarios conducive to synchrony. The first scenario (I) combines strong coupling with moderate detuning. The second scenario (II) combines moderate coupling with moderate detuning. Closed circles indicate two scenarios not conducive to synchrony. The third scenario (III) combines weak coupling with moderate detuning. The fourth scenario (IV) combines moderate coupling with large detuning. **b**, Translation of the four scenarios to stimulus features. Detuning and coupling strength map onto contrast heterogeneity and grid coarseness through the anatomy and physiology of early visual cortex. In this simplified illustration, two texture elements (Gabor annuli) fall within receptive fields of neural assemblies (purple) in early visual cortex. Contrast determines oscillation frequency (orange), with higher contrasts leading to higher frequencies. Differences in contrast within the receptive fields of two neural assemblies thus leads to differences in their frequencies and hence to higher detuning. Note that neural assemblies typically have several Gabor annuli in their receptive fields and extract their average contrast. There is thus no one-to-one mapping between annuli and receptive fields in our model. Coupling strength (line thickness of connecting arrow) depends on cortical distance, which due to retinotopy directly relates to the distance between texture elements in the visual field. Larger distances between annuli thus stimulate more remote neural assemblies with weaker coupling. **c**, Example full texture stimulus comprised of nonoverlapping Gabor annuli on irregular grid. For all participants and sessions 1-8, the lower right quadrant contains a vertical figure (magenta outline, not shown to participants). Blue dot: fixation point. Axes separating quadrants shown for illustration only, not visible to participants. On a given trial, the figure may be vertical or horizontal and participants indicated the figure's orientation. **d**, Figure region cut-outs illustrating experimental conditions. Grid coarseness (five steps) manipulates coupling strength for both figure and background. Contrast heterogeneity (five steps) manipulates detuning within figure. Background always at maximum heterogeneity (equivalent to rightmost column). The 25 cut-outs show all combinations of grid coarseness and contrast heterogeneity used in the experiments.

A synchrony-based grouping mechanism based on these principles has been successfully exploited for image segmentation in machine vision (Fang et al., 2014; Lowet et al., 2015; Nikonov et al., 2020). Here we bring these perspectives together to test whether human vision likewise behaves in accordance with TWCO principles. To test this hypothesis, we used a figure-ground segregation paradigm wherein human observers reported the orientation of a rectangular figure region in a texture stimulus composed of Gabor annuli (see Figure 1c for an illustration). According to TWCO, synchrony is governed by the interplay between oscillator detuning and coupling strength (Acebrón et al., 2005). Therefore, we created stimuli in which these two core parameters were systematically manipulated. We parametrically varied contrast heterogeneity as an implementation of frequency detuning and grid coarseness as an implementation of the cortical distance that determines coupling strength (see Figure 1d for an illustration). The figure was defined by a less heterogeneous contrast distribution between the elements, compared to elements in the background. Additionally, we investigated whether this synchrony-based grouping mechanism is adaptive by using a perceptual learning paradigm in which participants improved their perceptual performance over 8 daily sessions. By formalizing the principles of TWCO in a V1 oscillator model augmented with a simple Hebbian learning mechanism, we derived quantitative predictions from the theory. Our psychophysics results align well with the synchrony exhibited by the model, supporting the idea that stimulus-dependent gamma synchrony may be behaviorally relevant.

Results

Eight participants (6 female, mean age = 23.75, standard deviation = 6.453) performed a two-alternative forced choice texture discrimination task. We employed a repeated-measures design with extensive sampling. A design analysis indicated that our sample size afforded approximately 92% posterior detection probability (analogous to statistical power) for the core effects (Supplementary Table 1). Texture stimuli consisted of nonoverlapping Gabor annuli on an irregular grid (see Figure 1c). Each Gabor annulus was characterized by its own local contrast and was equiluminant with the background. Within a single visual quadrant, a rectangular figure was defined by less heterogeneity in the contrasts of local Gabor elements compared to the background, while keeping mean contrast between figure and background equal. Participants indicated the orientation (horizontal vs vertical) of the figure while fixating centrally. We manipulated two factors. The first was contrast heterogeneity within the figure, which we operationalized as the width of a uniform distribution from which annulus contrast values were drawn. This distribution was centered around a mean contrast of 50%. The background exhibited maximum contrast heterogeneity (from 0% to 100%). The second factor was the coarseness of the grid (distance between annuli). This manipulation affected figure and background equally. Both factors were manipulated in five steps resulting in 25 conditions (see Figure 1d). Within an experimental session, participants completed 30 blocks of each condition (750 trials). Participants received feedback after each trial in the form of color changes of the fixation point. Eye-tracking was used to ensure fixation, and trials where fixation was broken during either the fixation period preceding the stimulus, or during stimulus presentation, were aborted and repeated at a randomly chosen time later in the session. The experiment consisted of 9 consecutive sessions (8 training and 1 transfer session). In the transfer session, the rectangular figure was moved to the diagonally opposite quadrant.

To provide a mechanistic link between contrast heterogeneity, grid coarseness and synchrony in early visual cortex on the one hand, and quantitative predictions of discrimination accuracy on the other, we developed a phase-oscillator model of V1. The model represents a patch of visual space corresponding to the figure region in our psychophysics experiments, mapped onto V1 using a complex-logarithmic topographic transformation (Balasubramanian & Schwartz, 2002; Schwartz, 1980). To reduce computational cost, we only modeled the figure and not the background, under the assumption that the synchrony level in one image region would not be substantially altered by the synchrony level in the other image region (see Supplementary Figure 1). Based on this, synchrony in the background at maximum contrast disparity was equated to

synchrony in the figure at that contrast disparity. Each model oscillator represents a neural assembly receiving local input from the visual field. The frequency of each oscillator is a quasi-linear function of the contrast falling inside its receptive field, as has been determined previously in macaques (Evers et al., 2021 [↗](#); Roberts, Lowet, Brunet, TerWal, Tiesinga, Fries, & DeWeerd, 2013 [↗](#)). Receptive fields are modelled as isotropic 2D Gaussian functions with sizes that scale with eccentricity according to human cortical magnification (Freeman & Simoncelli, 2011 [↗](#)). Furthermore, we included recurrent connections between phase oscillators reflecting the lateral anatomical connectivity among columns in V1 and other low-level visual areas (Crist et al., 2001 [↗](#)). In line with anatomical data (Amir et al., 1993 [↗](#); Eckhorn, 1994 [↗](#); Gilbert & Wiesel, 1989 [↗](#); Ts'o et al., 1986 [↗](#)), coupling strength in our model declines exponentially with physical distance along the cortical surface. Our model captures this with two parameters estimated from independent neurophysiological data (Lowet, Roberts, Peter, Gips, & De Weerd, 2017 [↗](#)), Peter, Gips, & De Weerd, 2017): maximum coupling strength γ and coupling decay factor λ . The model was exposed to the same figure region texture stimuli as human participants, with manipulations of contrast heterogeneity and grid coarseness. We quantified the model's degree of zero-lag synchrony as the magnitude of the Kuramoto order parameter (synchronization index).

In our V1 model, learning is implemented to occur offline between simulated sessions, following a Hebbian-type learning rule that adapts coupling strengths based on pairwise phase-locking values (PLVs) accumulated over trials within a session. The contribution of each trial to learning is weighted by the probability of a correct response, determined by a psychometric function relating model synchrony to performance. This learning mechanism implies that connections between oscillators that exhibited coherence on correct trials are strengthened, bounded by the maximum coupling strength. Incorporating an upper bound on connections was motivated by findings that synaptic strength is limited by intrinsic properties of vesicular docking (Malagon et al., 2020 [↗](#)) and that late long-term potentiation approaches a maximum after several repeated experiences (Kandel et al., 2000 [↗](#)). Free parameters of the learning mechanism were estimated using data from the first two experimental sessions. To maximally disentangle data used for adjusting model parameters and data used for testing model predictions, we employed a leave-one-out cross-validation procedure. Model parameters were repeatedly estimated from the first two sessions in seven of our eight participants and the resulting model was used to predict performance in the remaining six sessions of the left-out participant. Our model rests on the assumption that learning-induced structural changes in early visual cortex are specific to the retinotopic locations of the trained stimuli. We evaluated whether this assumption holds for our human participants using the transfer session following the main training period. In the transfer session, participants performed the texture discrimination task with the figure region moved to a visual quadrant that had not been previously exposed to the figure. If learning is indeed local, participants' performance in the transfer session should resemble that of early training sessions, indicating a reset in performance for the new retinal location. On the other hand, if learning generalizes across retinal locations, performance in the transfer session should maintain the improvements seen in later training sessions. By comparing transfer session performance to both early and late training sessions, we can evaluate the validity of our model's assumption.

Synchrony Principles Govern Static Figure-Ground Perception

We first asked the question whether the factors that determine synchrony among coupled oscillators, frequency detuning and coupling strength, are predictive of the human ability to segregate a rectangular figure from its background in texture stimuli. In early visual cortex, oscillation frequency directly maps onto the contrast of texture elements (Hadjipapas et al., 2015 [↗](#); Lowet et al., 2015 [↗](#); Roberts, Lowet, Brunet, TerWal, Tiesinga, Fries, & DeWeerd, 2013 [↗](#)) and coupling strength directly maps onto their physical proximity (Gilbert & Wiesel, 1983 [↗](#); Lowet et al., 2015 [↗](#); Lowet, Roberts, Peter, Gips, & De Weerd, 2017 [↗](#); Stettler et al., 2002 [↗](#); Ts'o et al., 1986 [↗](#)). If texture segregation indeed depends on the synchrony principles identified by the theory of weakly coupled oscillators (TWCO), we expect discrimination accuracy to reveal a “behavioral” Arnold tongue in the space defined by contrast heterogeneity and grid coarseness.

To test these predictions, we analyzed the main effects of contrast heterogeneity and grid coarseness, as well as their interaction, on discrimination accuracy using Bayesian hierarchical logistic regression. This allowed us to analyze individual trial data rather than aggregated accuracy, while simultaneously accounting for within-subject variability by estimating participant-specific intercepts and slopes for each predictor. Both contrast heterogeneity and grid coarseness were z-normalized prior to fitting the statistical model. Note that while the principles of TWCO primarily predict main effects of contrast heterogeneity and grid coarseness, we additionally included their interaction to capture complex relationships specific to V1 that are not immediately apparent from the general theory. Specifically, coupling strength decays exponentially with cortical distance, which itself depends on cortical magnification. This should lead to a highly nonlinear relationship between grid coarseness and coupling strength that is likely to manifest as an interaction. In line with our expectations, the test provided strong evidence that both increased contrast heterogeneity ($\beta = -0.60$, 95% HDI $[-0.89, -0.30]$, $\text{Pr}[\beta < 0] = 0.999$, OR = 0.56, 95% HDI for OR $[0.41, 0.74]$) and grid coarseness ($\beta = -0.27$, 95% HDI $[-0.40, -0.13]$, $\text{Pr}[\beta < 0] = 0.999$, OR = 0.77, 95% HDI for OR $[0.67, 0.88]$) reduced discrimination accuracy (Figure 2a,b). These results provide credible evidence that a one-standard-deviation increase in contrast heterogeneity reduces the odds of a correct response by approximately 44%, while a similar increase in grid coarseness reduces the odds by 23%. Furthermore, Figure 2a,b shows a behavioral Arnold tongue as a triangular region of high accuracy ($\geq 75\%$ correct). There was likewise strong evidence for an interaction between contrast heterogeneity and grid coarseness ($\beta = 0.24$, 95% HDI $[0.12, 0.36]$, $\text{Pr}[\beta > 0] = 0.998$, OR = 1.27, 95% HDI for OR $[1.12, 1.44]$). This indicates that the specific characteristics of early visual cortex contribute beyond the general principles of TWCO.

Our model of V1 captures both the general principles of TWCO as well as idiosyncratic characteristics of early visual cortex in a single mechanism, and we expected this model to predict the human ability to segregate a rectangular figure from its background in texture stimuli. Indeed, the synchrony exhibited by our model (Figure 2c,d), when exposed to the same stimuli as our participants, resembled behavioral discrimination accuracy (Figure 2a,b). A Bayesian hierarchical logistic regression with model synchrony as sole predictor revealed strong evidence that it is associated with improved accuracy ($\beta = 0.76$, 95% HDI $[0.33, 1.21]$, $\text{Pr}[\beta > 0] = 0.998$, OR = 2.19, 95% HDI for OR $[1.38, 3.33]$). This represents credible evidence that a one-standard-deviation increase in synchrony more than doubles the odds of a correct response. Hence, our proposed mechanism is capable of reproducing the key patterns in the behavioral data.

A natural question is whether synchrony constitutes the unique mechanistic link from stimulus features to perception within our model. To address this question, additional analyses used the average model firing rates within the figure as a predictor for segregation, as well as the difference between average model firing rates inside and outside the figure. The latter rate difference between figure and ground can serve as a phenomenological proxy for putative rate-based segregation mechanisms. Note that we treat the instantaneous frequency of each oscillator as a proxy for the instantaneous population firing rate of the corresponding neural assembly.

With respect to the average figure firing rates, we found some evidence indicating that they were associated with segregation accuracy ($\beta = 0.07$, 95% HDI $[-0.025, 0.16]$, $\text{Pr}[\beta > 0] = 0.941$, OR = 1.07, 95% HDI for OR $[0.98, 1.18]$). However, because the 95% highest density interval included zero, we evaluated whether the effect fell within a Region Of Practical Equivalence (ROPE) of $\pm 2\%$ accuracy and found only weak evidence for this ($\text{Pr}[|\Delta \text{acc}| < 0.02] = 0.680$). Hence, the effect is likely present, but small. With respect to rate differences, we found credible evidence that they could be associated with accuracy ($\beta = -0.55$, 95% HDI $[-0.78, -0.32]$, $\text{Pr}[\beta < 0] = 0.999$, OR = 0.58, 95% HDI for OR $[0.46, 0.73]$). However, the effect of rate difference was smaller than that of synchrony. Furthermore, firing in the figure was reduced compared to background firing.

We next compared synchrony, average figure firing rate and rate differences derived from the same V1 simulations in terms of their out-of-sample predictive accuracy using Pareto-smoothed importance sampling leave-one-out cross-validation. Synchrony was favored over rate difference ($\Delta \text{ELPD} \approx 19$, $\text{dSE} \approx 14$) and average figure firing ($\Delta \text{ELPD} \approx 127$, $\text{dSE} \approx 17$). The stacking weights further support this with synchrony receiving a weight of 0.90, rate difference a weight of 0.10,

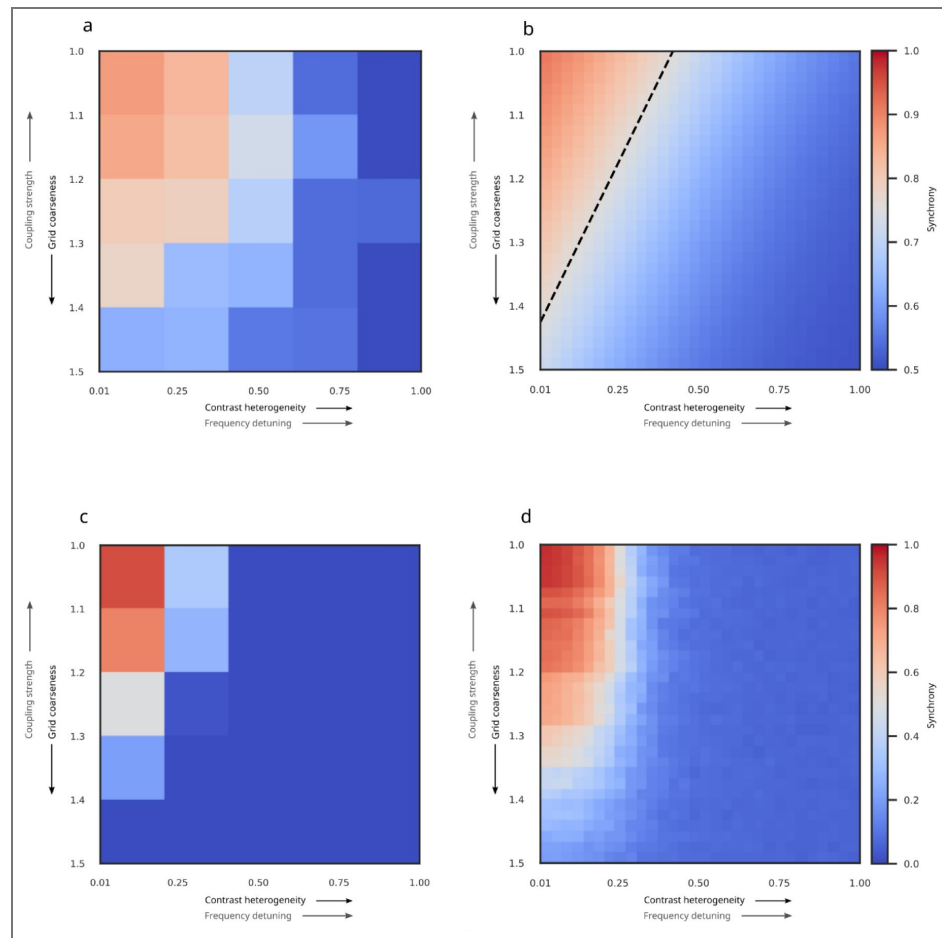


Figure 2. Behavioral and simulated Arnold tongues.

a, Average discrimination accuracy for each of the 25 experimental conditions revealed a behavioral Arnold tongue in the space defined by contrast heterogeneity and grid coarseness. Contrast heterogeneity translates into the variance of frequencies (detuning) whereas grid coarseness translates into cortical distance (coupling strength). **b**, Fitted behavioral Arnold tongue after fitting a two-dimensional psychometric curve to the results in (a). The dashed line indicates the combination of contrast heterogeneity and grid coarseness corresponding to 75% accuracy. **c**, Zero-lag synchrony among model oscillators showing an Arnold tongue in the same parameter space as (a). Simulation conditions matched the 25 experimental conditions. **d**, High-resolution visualization of zero-lag synchrony, using 900 conditions (30 levels each of contrast heterogeneity and grid coarseness) to provide a more detailed representation of the Arnold tongue.

and the average figure firing rates a weight of effectively zero. These results indicate that, for our stimuli and our V1 model, a synchrony-based readout provides the most faithful mapping from stimulus to perception among simple alternatives. However, this comparison does not rule out that more sophisticated rate-based models could provide viable mechanistic accounts of figure-ground segregation. Nevertheless, our data indicate that synchrony-based mechanisms are eminently viable.

A key strength of our model is that it does not depend on fine-tuning parameters to our behavioral data. To demonstrate this, we conducted a parameter space exploration of key choices of model parameter values (maximum coupling strength and coupling decay factor) and found that our choices, which were obtained from independent observations in macaques (Lowet, Roberts, Peter, Gips, & De Weerd, 2017) were already close to optimal. We used Pearson correlations (Figure 3a) and weighted Jaccard similarity (Figure 3b) to assess the similarity between the behavioral Arnold tongue and the Arnold tongue predicted by our V1 model for various combinations of maximum coupling strength and coupling decay factor. We included both correlations and Jaccard similarity because the former is more widely known while the latter is more conservative. To compute weighted Jaccard similarity between two sets of real numbers, they need to fall within the same range. Accordingly, we applied min-max normalization to ensure that discrimination accuracy fell within a zero-to-one range matching the range of the synchronization index. This procedure yielded the similarity comparisons color-coded in Figure 3. The point marked with the black dot reflects the parameter value combination that was based on independent macaque data (24.63 and 0.22, respectively) and that was exclusively used for our model predictions. When using Pearson correlations as a similarity measure, this parameter value combination fell just within the region of optimal parameter values for our behavioral results (Figure 3a). When using the more conservative weighted Jaccard similarity index (Figure 3b), our chosen parameter value combination appeared slightly outside of the optimal region. Thus, the two model parameters estimated from neurophysiological recordings in monkeys were close to optimal for predicting human perceptual behavior, but not fully optimal. This may be due to horizontal connections in human visual cortex extending further than those in the macaque (Amir et al., 1993; Burkhalter & Bernardo, 1989; Lund, Yoshioka, & Levitt, 1993; Voges et al., 2010; Yoshioka et al., 1996), suggesting a slightly smaller coupling decay factor in humans. Applying a smaller coupling decay would move model parameters into the optimal regime, thereby extending the predicted Arnold tongue (Figure 2c) diagonally in the direction of the behavioral Arnold tongue (Figure 2a). The parameters maximum coupling and decay factor might reflect biological constraints on the strength of lateral connections (Kandel et al., 2000; Malagon et al., 2020; Rioult-Pedotti et al., 1998), which to some extent may differ between monkeys and humans.

Plasticity-Induced Changes in Synchrony Quantitatively Predict Perceptual Learning

Our results show that a neural grouping mechanism based on synchrony principles can account for behavioral performance, suggesting it is a viable candidate for explaining texture segregation. We next asked whether training-induced changes of lateral connections among neural assemblies in early visual cortex affect assemblies' readiness to synchronize and whether this is accompanied by performance improvements. We reasoned that neural synchrony must remain adaptable to the statistics of visual experiences to function effectively as a grouping mechanism. Consequently, we hypothesized that if synchrony among neural assemblies is related to figure-ground segregation and enhanced through perceptual learning, the ability to segregate figure from ground should increase with training.

To test this, both the model and human participants were exposed to eight daily sessions of extensive training using identical stimuli and experimental conditions. We hypothesized that both grid coarseness and contrast heterogeneity exhibit main effects on discrimination accuracy.

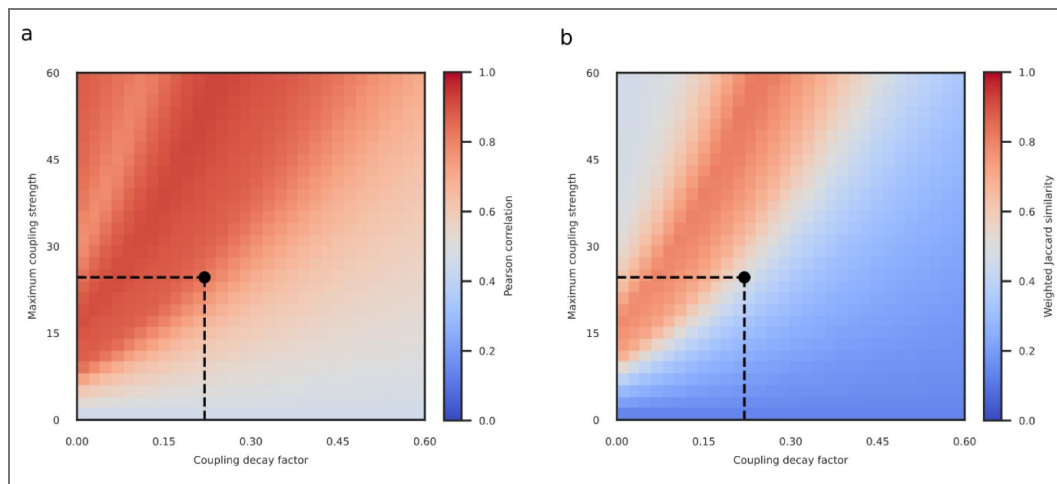


Figure 3. Comparison of behavioral and simulated Arnold tongues across coupling parameter space.

a, Pearson correlation between the behavioral Arnold tongue and simulated Arnold tongues obtained from models with coupling weights determined by different combinations of maximum coupling strength and coupling decay factor. The point labelled by the black circle shows the combination of parameters that were obtained from independent (macaque) data. **b**, Weighted Jaccard similarity between the behavioral Arnold tongue and simulated Arnold tongues. This metric is displayed across the same parameter space as in (a).

However, we also expected coupling strength to increase with learning and that this would allow synchrony to occur for increasingly coarser grids. We thus hypothesized an interaction effect between session and grid coarseness on discrimination accuracy. Furthermore, we hypothesized an additional interaction between session and contrast heterogeneity where the effect of contrast heterogeneity would increase over sessions. Model simulations for the first session never revealed a synchronized state for contrast heterogeneity values beyond 0.25, even for the densest grids (see Figure 2c,d). This, together with an upper bound on coupling strength, suggested that synchrony cannot be achieved far beyond this cutoff point, even after extensive training. Indeed, model simulations of training confirmed this, showing that synchrony approached this cutoff point for increasingly coarser grids over sessions (Figure 4c). These model results indicate that the effect of contrast heterogeneity would increase over sessions with high performance for values below the cutoff point and low performance above the cutoff point. Finally, extensive training may globally increase participants' performance, implying a main effect of session.

To evaluate these predictions, we performed Bayesian hierarchical logistic regression including the main effects of contrast heterogeneity, grid coarseness, and session, as well as interactions between session and contrast heterogeneity, between session and grid coarseness, and between contrast heterogeneity and grid coarseness. As before, we included the interaction between contrast heterogeneity and grid coarseness to account for potential nonlinear effects specific to V1. The analysis revealed strong evidence that participants' ability to segregate figure from ground increased over sessions ($\beta = 0.095$, 95% HDI [0.066, 0.126], $\text{Pr}[\beta > 0] = 1.00$, OR = 1.10, 95% HDI for OR = [1.07, 1.13]). Furthermore, we found strong evidence for an interaction between contrast heterogeneity and session ($\beta = -0.081$, 95% HDI [-0.090, -0.070], $\text{Pr}[\beta < 0] = 1.00$, OR = 0.92, 95% HDI for OR = [0.91, 0.93]). We also found evidence for a small interaction between grid coarseness and session ($\beta = -0.008$, 95% HDI [-0.018, 0.001], $\text{Pr}[\beta < 0] = 0.954$, OR = 0.99, 95% HDI for OR = [0.98, 1.00]). However, the 95% highest density interval included zero. We subsequently confirmed that the change was within a region of practical equivalence (ROPE) of $\pm 2\%$ accuracy $\text{Pr}[|\Delta\text{acc}| < 0.02] = 0.999$. While the interaction between session and grid coarseness is thus negligible, there was strong evidence for a main effect of grid coarseness on discrimination accuracy ($\beta = -0.316$, 95% HDI [-0.377, -0.262], $\text{Pr}[\beta < 0] = 1.00$, OR = 0.73, 95% HDI for OR = [0.69, 0.77]) with increasing accuracy as grid coarseness decreased. As can be appreciated from Figure 4, there indeed seemed to be a cutoff value for contrast heterogeneity beyond which the figure could not be discriminated from the background. This cutoff may also explain why the interaction between session and grid coarseness was negligible. Below the cutoff point, the top and middle rows of Figure 4 suggest that participants could discriminate the figure for increasingly coarser grids. Beyond the cutoff, however, grid coarseness seemed to have had no discernible effect regardless of how much training participants received. The characteristic triangular shape of the Arnold tongue thus gradually morphed into a rectangular shape.

Next, we examined simple effects of contrast heterogeneity on discrimination accuracy for each session separately (see Table 1). As expected from the presence of the cutoff, the effect of contrast heterogeneity increased over sessions, reflected in decreasing log-odds (β) and corresponding odds ratios (ORs) over sessions as shown in Table 1.

Finally, we evaluated whether the observed effects reflected localized learning in early visual cortex, as assumed by our model, implying that the training effect would be specific to the trained location. Performance in the transfer session should thus resemble that observed at training locations during early rather than late sessions. To test this, we estimated a hierarchical Bayesian logistic regression model with predictors for contrast heterogeneity, grid coarseness, session, their interactions, and an indicator for the transfer session. Subject-level random intercepts and slopes were included. From the fitted model, we generated posterior predictions of the population-level mean accuracy for each session. We then compared transfer (Session 9) with an early reference session in two complementary ways. First, we estimated the posterior probability that transfer session accuracy was lower than in the reference session. Second, we estimated the posterior probability that the difference between accuracy in the transfer and reference session lay within a region of practical equivalence (ROPE, $\pm 2\%$ accuracy). We used the second session, the earliest

Figure 4. Learning effects on Arnold tongues.

a, Group average behavioral Arnold tongues for the 25 experimental conditions for each session. The vertical black line separates transfer session 9 from training sessions 1 to 8. **b**, Two-dimensional psychometric curves fitted to session-specific group average behavioral Arnold tongues. The dashed line again indicates the combination of contrast heterogeneity and grid coarseness at which participants achieve 75% accuracy. **c**, Simulated Arnold tongues for each of the eight training sessions including session-by-session learning in the model. We did not include a simulation of the ninth session because the location-specificity of the model learning rule would render it identical to the first session. Note that for visualization purposes we simulated the model for 30 levels of contrast heterogeneity and 30 levels of grid coarseness, in both cases including the 5 levels investigated experimentally.

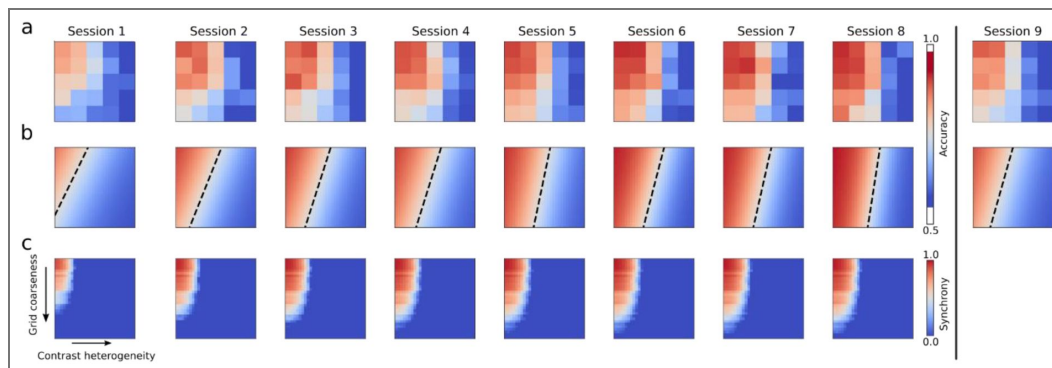


Table 1. Effects of contrast heterogeneity on discrimination accuracy across sessions.

HDI = Highest Density Interval. Log-odds and Odds Ratios represent the effect of a one standard deviation increase in contrast heterogeneity on the odds of correct discrimination. The probabilities of negative log-odds are for the simple effect of contrast heterogeneity in each session.

Session	Log-odds (β)	Odds Ratio (OR)	95% HDI for OR	$\Pr[\beta < 0]$
1	-0.66	0.52	[0.38, 0.66]	1.00
2	-0.74	0.48	[0.35, 0.61]	1.00
3	-0.82	0.44	[0.32, 0.56]	1.00
4	-0.90	0.41	[0.30, 0.52]	1.00
5	-0.98	0.38	[0.28, 0.48]	1.00
6	-1.06	0.34	[0.26, 0.44]	1.00
7	-1.14	0.32	[0.24, 0.41]	1.00
8	-1.22	0.30	[0.22, 0.38]	1.00

session after task familiarization, as reference. Our analysis revealed that performance in the transfer session was practically equivalent to session 2 (93% posterior probability of equivalence). Based on this, we expected that the local learning mechanism implemented in our model can provide quantitative predictions of performance changes over the course of the eight training sessions.

We evaluated the quantitative agreement between model synchrony and empirical discrimination performance. This analysis focused exclusively on synchrony. As we show in the supplementary materials (Suppl. Figure 2 [↗](#)), rate-based readouts of our V1 model are not at all affected by coupling strength. As such, they are insensitive to changes in coupling and are thus not viable as alternative mechanisms to explain performance changes due to learning. To evaluate the quantitative agreement between model synchrony and empirical discrimination performance, we measured the similarity between simulated and behavioral Arnold tongues using Pearson correlations and weighted Jaccard similarity. Because we employed a leave-one-out cross-validation procedure, we obtained eight simulated Arnold tongues in sessions 2-8 after optimizing learning parameters on data from seven participants. Simulated Arnold tongues in each fold were always compared to behavioral Arnold tongues of the left-out participant. The first session did not involve learning and model simulations were identical to those reported above. Note that data from the second session was used to adjust model parameters and hence only sessions 3-8 could be used for evaluating model predictions. This cross-validation approach enabled us to assess the model's ability to predict performance in unseen data, rather than merely fitting observed results post-hoc. Figure 5a [↗](#) and 5b [↗](#) show correlations and Jaccard similarity between simulated and behavioral Arnold tongues, respectively. Grey regions indicate a noise ceiling that was obtained by computing the fit between average behavioral Arnold tongues in a fold and the behavioral Arnold tongue of the left-out participant. The grey region marks the 25th to the 75th percentile of fit values obtained using this procedure. The figure demonstrates consistent quantitative agreement between simulated and behavioral Arnold tongues across sessions.

To examine this further, we tested whether the size of the simulated Arnold tongue across sessions was predictive of the size of the behavioral Arnold tongues. We quantified the size of each Arnold tongue in terms of the volume under its surface computed using Simpson's numerical integration. Arnold tongues grew across sessions with comparable growth curves for simulated and behavioral Arnold tongues (see Figure 5c [↗](#)). The precise relationship between simulated and behavioral Arnold tongue sizes is depicted in Figure 5d [↗](#). Subsequently, we performed a Bayesian hierarchical linear regression to investigate this in sessions 3-8. We ignored the first two sessions since these were used for estimating model learning parameters. As expected, the size of the simulated Arnold tongue predicted the size of the behavioral Arnold tongue ($\beta = 0.54$, 95% HDI [0.106, 0.935], $\text{Pr}[\beta > 0] = 0.992$). The model's capability to accurately reflect learning effects observed in human participants is consistent with the notion that enhanced synchrony among neural assemblies in early visual cortex resulting from perceptual learning enhances human's ability to segregate figure from ground. This further strengthens the view that synchrony principles provide a viable neural grouping mechanism for texture segregation.

Discussion

The role of synchrony in the gamma frequency band for visual perception remains a matter of debate (Duecker et al., 2021 [↗](#); Fernandez-Ruiz et al., 2023 [↗](#); Ray & Maunsell, 2015 [↗](#); Roelfsema, 2023 [↗](#)). A putative role for gamma synchrony in processing the features of a stimulus both within and across visual areas (Fries, 2009 [↗](#); Singer, 1999 [↗](#); Uhlhaas et al., 2008 [↗](#); Womelsdorf et al., 2007 [↗](#)) has been called into question based on the stimulus-dependence of gamma synchrony (Ray & Maunsell, 2010 [↗](#), 2015 [↗](#); Roelfsema, 2023 [↗](#)). Alternatively, it has also been suggested that feature-dependent gamma frequencies and distance-dependent synchrony are key ingredients in a neural grouping mechanism underlying figure-ground segregation (Lowet et al., 2015 [↗](#); Lowet, Roberts, Peter, Gips, & De Weerd, 2017 [↗](#)). It is well-established that the frequency of gamma oscillations in visual cortex depends on local stimulus features (Baldi & Meir, 1990 [↗](#); Buia & Tiesinga, 2006 [↗](#); Hall et al., 2005 [↗](#); Henrie & Shapley, 2005 [↗](#); Roberts, Lowet, Brunet, TerWal,

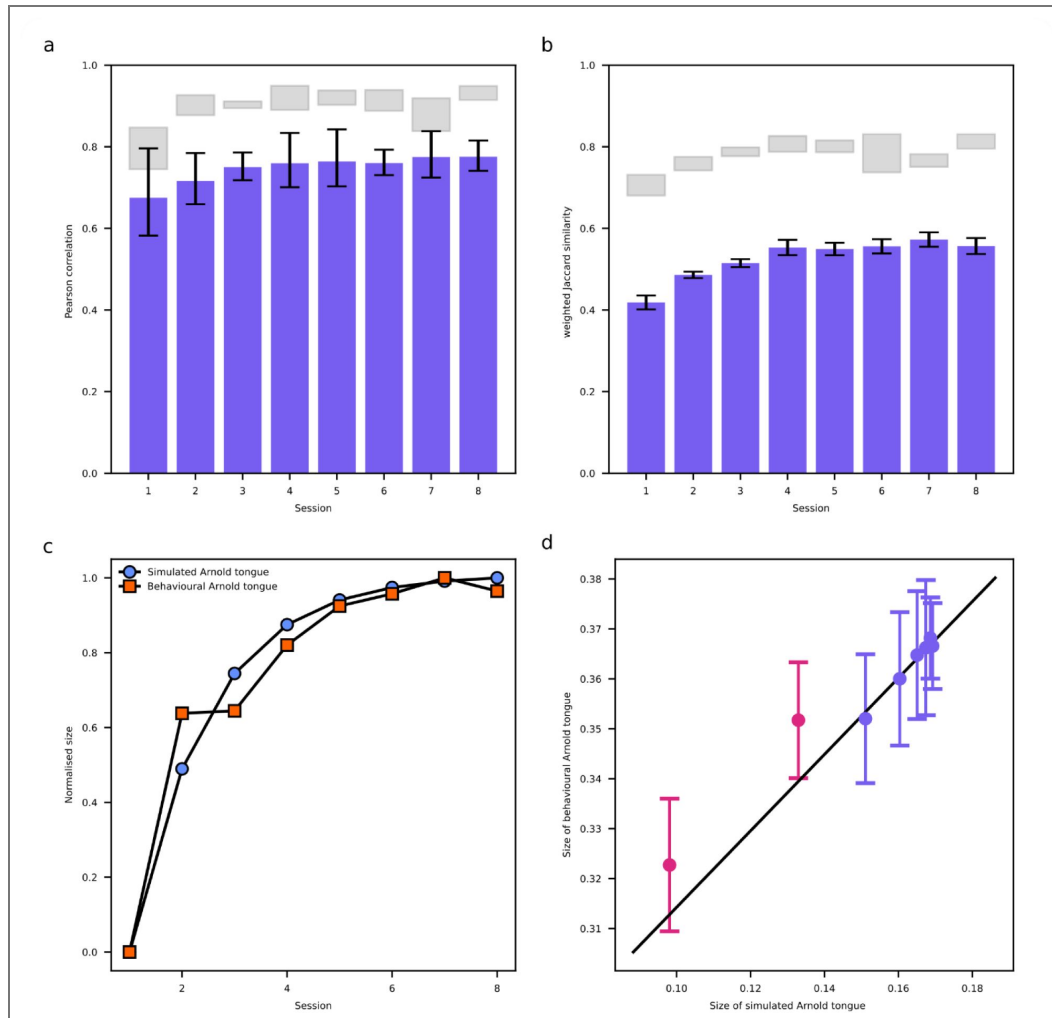


Figure 5. Model predictions of learning effects.

a, Pearson correlations between simulated and behavioral Arnold tongues for each training session. Error bars indicate 95% confidence intervals. Grey regions indicate a noise ceiling that was obtained by computing the fit between average behavioral Arnold tongues in a fold and the behavioral Arnold tongue of the left-out participant. The grey regions reflect the 25th to the 75th percentile of fit values obtained using this procedure. **b**, Weighted Jaccard similarity values between simulated and behavioral Arnold tongues for each training session. Error bars and grey regions as in (a). **c**, Sizes of simulated (blue circles) and behavioral (orange squares) Arnold tongues across sessions. Arnold tongue sizes were averaged across participants and subsequently min-max normalized. This normalization highlights the growth patterns while accounting for the different value ranges of simulated and behavioral Arnold tongues. **d**, Sizes of behavioral Arnold tongues as a function of sizes of simulated Arnold tongues. The best fitting regression (black line) was obtained from a mixed effects model fitted to data from sessions 3-8 (blue circles). Red circles reflect data from the first two sessions that was not included in the mixed effect model. The black line was extended to include these points. Error bars indicate 95% confidence intervals.

Tiesinga, Fries, & DeWeerd, 2013 [↗](#); Shapira et al., 2017 [↗](#)) and that lateral connectivity between neural groups within early visual cortex depends on cortical distance (Amir et al., 1993 [↗](#); Boucsein et al., 2011 [↗](#); Eckhorn, 1994 [↗](#); Gilbert & Wiesel, 1989 [↗](#); Stettler et al., 2002 [↗](#); Ts'o et al., 1986 [↗](#)). It is likewise a well-known property of coupled oscillators that they synchronize when their coupling is sufficiently strong to overcome differences in their frequency, but not otherwise (Acebrón et al., 2005 [↗](#); Ermentrout et al., 2019 [↗](#); Kuramoto, 1984 [↗](#); Neu, 1979 [↗](#); Strogatz, 2000 [↗](#)). Synchrony may thus drive the perceptual grouping of elements if they are sufficiently similar to each other within one image region, and thereby segregate it from other image regions based on their different levels of synchrony.

We tested this hypothesis in a psychophysics experiment wherein human observers discriminated the orientation of a texture-defined, rectangular figure region (vertical vs horizontal). The stimulus consisted of small Gabor annuli arranged on an irregular grid. Each Gabor annulus was characterized by its own contrast and the figure region was defined by less heterogeneous contrasts among the Gabor annuli compared to the background. We manipulated contrast heterogeneity and grid coarseness (distance between annuli) as a proxy of frequency detuning and coupling strength, respectively. Both contrast heterogeneity and grid coarseness affected discrimination accuracy. Specifically, we found that accuracies beyond 75% were limited to a triangular region in the space spanned by these two factors, forming a behavioral Arnold tongue. In line with our expectations, increased contrast heterogeneity in the figure permitted figure-ground segregation if accompanied by a reduction in grid coarseness. These results quantitatively aligned well with synchrony exhibited in a coupled oscillator V1 model exposed to the same texture stimuli.

The capacity of our model to predict human psychophysical performance is notable given that the key parameters of maximum coupling strength and coupling decay factor were obtained from neurophysiological data recorded from macaques (Lowet, Roberts, Peter, Gips, & De Weerd, 2017 [↗](#)). This cross-species validation underscores the robustness of our model and suggests that the neural mechanisms underlying gamma oscillations and figure-ground segregation are largely conserved across primate species (Buzsáki et al., 2013 [↗](#)). It is noteworthy that the parameter combination obtained from macaque data bordered the region of optimal combinations exhibiting the highest match to psychophysics results that our model could, in principle, achieve (see Figure 3 [↗](#)). The slight deviation from the optimal regime likely stems from the fact that parameters were estimated from data obtained in macaques and subsequently used to predict human behavior. It is likely that horizontal connections in the human extend further than those in the macaque (Amir et al., 1993 [↗](#); Burkhalter & Bernardo, 1989 [↗](#); Lund, Yoshioka, & Levitt, 1993 [↗](#); Lund, Yoshioka, Lund, et al., 1993 [↗](#); Voges et al., 2010 [↗](#)) and may thus be associated with a slightly smaller coupling decay factor. Another possibility is that the parameter value we derived from Lowet et al. (2017) [↗](#), a study chosen because their paradigm targets the same TWCO components that guided our stimulus design, is an overestimate. As with any study, their data comes with uncertainty such that our estimates might not perfectly reflect actual decay rates. While we currently do not have alternative data to estimate the exact human decay factor and hence cannot establish how much model fit would be affected, any small to modest reduction would certainly further improve model fit.

To further investigate whether synchrony among neuronal populations exhibiting contrast dependent frequencies provides a potential perceptual grouping mechanism, we tested whether training-induced changes of lateral coupling in a network of phase oscillators improved the readiness of these oscillators to synchronize and whether this model provided accurate predictions of performance on the figure-ground segregation task. We reasoned that for neural synchrony to function effectively as a grouping mechanism, it should be modifiable by experience in a manner that matches training-induced improvements in texture segregation. We observed that discrimination performance improved as a function of training session, in line with participants' growing experience with the stimuli. Importantly, we found that training-induced increases in accuracy were well accounted for by model predictions of synchrony strength inside the figure. Our results are consistent with the notion that synchrony mechanisms in low-level visual areas contribute in a behaviorally relevant manner to texture segregation and that training-

induced changes of local synchrony are reflected by concurrent changes in perception. Synchrony and discrimination accuracy revealed highly congruent Arnold tongues. A close quantitative resemblance of these Arnold tongues was maintained across sessions as both tongues grew and changed form in a highly consistent manner. This supports the idea that learning-induced changes in figure-ground segregation may be mediated by plasticity-induced changes in synchrony. Oscillations have been shown to facilitate learning through spike-timing dependent plasticity (Masquelier et al., 2009 [↗](#)), rendering an oscillation-based Hebbian learning mechanism biologically plausible.

It is important to note that the learning mechanism integrated into our model assumes that learning is local. We validated this assumption in the human participants by testing whether moving the figure region from its trained location to a new location would lead to transfer of performance to the new location, or rather a decrease in performance in the new location. Our results supported the latter. This is in line with other studies that demonstrated that after location-specific training, low-level visual areas contribute to the location and stimulus specificity of expert visual performance (Karni & Bertini, 1997 [↗](#)). Based on previous findings that location-specific training induces localized plasticity in low-level visual areas (Brosch et al., 2015 [↗](#); Raiguel et al., 2006 [↗](#); Schoups et al., 2001 [↗](#); Yang & Maunsell, 2004 [↗](#)), we further assumed that learning in our paradigm primarily affects lateral connectivity within V1 and hence manipulates coupling strength between neural assemblies. An alternative hypothesis could be that learning, by targeting feedforward or feedback connectivity, alters the contrast sensitivity of neural assemblies. If this were to reduce the slope of the contrast-frequency relationship, it could theoretically offer a pathway to achieve synchrony across more heterogeneous contrasts by minimizing detuning rather than increasing coupling strength. However, empirical evidence suggests that training on perceptual tasks tends to steepen, rather than flatten, the contrast-frequency relationship (Chen et al., 2013 [↗](#); Hua et al., 2010 [↗](#); Sanayei et al., 2018 [↗](#)). Given its lack of empirical support, we therefore did not incorporate this alternative mechanism into our model.

The predominant cue for figure-ground segregation in our stimuli lay in the global variations of population statistics in the contrast distribution, rather than local differences at the boundary between the figure and the ground (de Weerd et al., 1994 [↗](#); Poort et al., 2016 [↗](#); Roelfsema et al., 2002 [↗](#)). This design was specifically chosen to preclude simple segregation based on mean firing rates. Nevertheless, our results indicate that a region comparison mechanism could still exploit firing rates to segregate figure from ground. While the firing rates of individual oscillators in our model are modulated by ongoing interactions, average firing rates in the figure region are insensitive to these interactions and hence purely driven by feedforward contrast extraction. By contrast, synchrony in our model arises from these interactions as they convert variance in local firing rates into coherence signals. Our results show that this can provide sufficient information for a subsequent read-out mechanism to distinguish figure from background. It might also provide additional information that downstream regions might exploit in addition to information carried by average firing rates. It might, for instance, provide a scaffold that can then be refined and read out by top-down mechanisms (Ahissar & Hochstein, 1997 [↗](#), 2004 [↗](#); Hochstein & Ahissar, 2002 [↗](#); Liu & Weinshall, 2000 [↗](#); Rubin et al., 1997 [↗](#)). Such a scaffold might be compatible with widely accepted recurrent models in which boundary detection is followed by region-filling feedback (Grossberg & Mingolla, 1985 [↗](#), 1987 [↗](#); Keil et al., 2005 [↗](#); Layton et al., 2014 [↗](#); Motoyoshi, 1999 [↗](#); Neumann et al., 2001 [↗](#); Pessoa & de Weerd, 2003 [↗](#); Roelfsema et al., 2002 [↗](#)), a notion substantiated by neurophysiological and psychophysical evidence (Poort et al., 2016 [↗](#); Roelfsema et al., 2002 [↗](#); Self et al., 2012 [↗](#)) as well as by lesion and optogenetics experiments (Kirchberger et al., 2021 [↗](#); Lamme et al., 1998 [↗](#); Supér & Lamme, 2007 [↗](#)). We must note that we used the instantaneous frequencies of our model oscillators as a proxy for population firing rates. This is an oversimplification given that population firing rates are much lower than gamma (Zachariou et al., 2021 [↗](#)). However, over the contrast range relevant to our stimuli gamma frequency and population firing co-vary approximately linearly (Zachariou et al., 2021 [↗](#)). Frequency thus served as a rate-like activation measure rather than a literal firing rate. It is furthermore important to acknowledge that our model does not account for attentional effects, although the significance of attention in figure-ground segregation and in learning is well-established (Huang et al., 2020 [↗](#)).

and it is likely that pure exposure to the stimuli in our experiment would have revealed very limited effects (Seitz & Dinse, 2007 [↗](#)). Thus, while the current model indicates what early visual circuits could achieve in isolation, integrating the synchrony scaffold with rate-based mechanisms and attentional gain control remains a goal for future work.

A consideration to keep in mind in interpreting the effects of training on texture segregation is that participants at the outset of the experiment were unfamiliar with various aspects of the task unrelated to the perceptual challenge itself. They had to learn to maintain fixation, to establish stimulus-response mappings and associated decision processes, in addition to solving the perceptual challenge. As such, results in the first session may represent cognitive processes related to these non-perceptual factors. Future versions of our experiment might consider including a baseline training session during which participants get acquainted with the experimental setup and task using stimuli that define figure and background with features that are independent of those manipulated in the main experiment. Moreover, participants were not informed of which visual quadrant the figure would appear in the transfer session. This raises the concern that our results partly reflect visual search effects (Eckstein, 2011 [↗](#); Neisser, 1964 [↗](#)) rather than a return to a naïve state of the figure-ground segregation skill. Arguably, however, this only affected a few trials and is thus insufficient to account for the loss of skill we observed. Furthermore, our model was designed to test the emergence of synchrony within the figure region itself, and as such, it did not include the background texture. While this approach allowed us to isolate the core mechanism of interest, it means our model provides an account of local grouping rather than a full simulation of figure-ground segregation. Finally, although a strength of this work is the prediction of human psychophysical performance based on a model whose parameters were set by independent neurophysiological data, a weakness is the absence of neurophysiological data for our specific experimental paradigm. Such data would allow for a full mediation analysis from stimulus features via synchrony to behavior and could strengthen our interpretations. At the same time, a combined psychophysical and neurophysiological experiment in an animal model replicating the experimental conditions used here would benefit from strong predictions provided by the present study as well as prior neurophysiological data (Lowet, Roberts, Peter, Gips, & De Weerd, 2017 [↗](#)). Despite these considerations and limitations, our results support the notion that gamma synchrony can serve a mechanistic role in figure-ground segregation.

The synchrony-based grouping mechanism studied here provides a theoretical framework for previous experimental results. A wide range of texture manipulations have been shown to drive segregation, including contrast (Hadjipapas et al., 2015), spatial frequency (Bredfeldt & Ringach, 2002; Henriksson et al., 2008), color (Shapley & Hawken, 2011), orientation (Lamme, 1995) and movement direction (Lamme, 1995). It is well documented that the difference between figure and background in one or a combination of these features (Landy & Bergen, 1991; Motoyoshi & Nishida, 2001; Nothdurft, 1985a, 1991b, 1991a) in population statistics (de Weerd et al., 1992; Nothdurft, 1985b) and in the physical proximity among texture elements within a figure (de Weerd et al., 1992; Nothdurft, 1985b) are the main parameters that determine the accuracy of figure-ground segregation. Much of this work consists of separate studies focusing on the contributions of single or restricted subsets of features to segregation. Viewed through the lens of TWCO, however, these features have their effect through the same mechanism. Most element features directly influence frequency detuning (Dubey & Ray, 2020; Hadjipapas et al., 2015; Henrie & Shapley, 2005; Roberts, Lowet, Brunet, TerWal, Tiesinga, Fries, & DeWeerd, 2013; Shapira et al., 2017), while proximity determines coupling strength via lateral connectivity in early visual cortex (Boucsein et al., 2011; Gilbert & Wiesel, 1983; Lowet et al., 2015; Lowet, Roberts, Peter, Gips, & De Weerd, 2017; Stettler et al., 2002; Ts'o et al., 1986). Rather than introducing a new explanatory variable, TWCO offers a mechanistic synthesis and shows how the established influence of these features on perception can emerge from the dynamics of coupled neural oscillators in V1. Thus, the success of these manipulations may arise precisely because they tap into the factors that determine whether synchrony can form among neural assemblies. As such, TWCO may provide a unifying principle that explains why these stimulus features are effective in modulating the efficiency of figure-ground segregation.

Future work should explore to what extent the principles of TWCO can explain segmentation of objects in natural images. While synchrony-based grouping mechanisms based on these principles have been used to segment natural images in machine vision (Fang et al., 2014; Lowet et al., 2015; Nikonov et al., 2020), it remains an open question whether cortical synchrony mediates human perception for such stimuli. Similarly, it remains an open question whether the principles outlined here generalize beyond the visual system to other sensory modalities. Interestingly, related forms of stimulus-dependent synchrony have been observed in auditory cortex, where it facilitates the integration of sound features and the segregation of auditory streams (Giraud & Poeppel, 2012). Finally, the principles of TWCO might provide a novel lens through which we can understand perceptual symptoms in neurological and psychiatric disorders. For example, schizophrenia is characterized by disrupted perceptual grouping and figure-ground segregation (Liddle, 1987; Malaspina et al., 2004; Uhlhaas et al., 2006) and disrupted visual gamma (Spencer et al., 2003). The prominent role of coupling strength within TWCO raises the possibility that reduced dendritic spine density in layer 3 of V1 within schizophrenia patients (Fish et al., 2025) may contribute to disrupted gamma synchrony and that this, in turn, may lead to disrupted perceptual grouping.

In conclusion, this study shows that figure-ground segregation performance can be well predicted by the factors that determine synchrony according to the theory of weakly coupled oscillators. Frequency detuning driven by contrast heterogeneity and coupling strength driven by physical distance may interact constructively to give rise to the perceptual skill of figure-ground segregation as well as its practice-induced enhancement. Our results show that a synchrony-based neural grouping mechanism can account for the observed behavioral patterns in a texture segregation task, and therefore remains a viable explanation for figure-ground segregation that cannot be ruled out. The documented dependence of gamma synchrony on stimulus features and element distance are essential components rather than obstacles to such a mechanism. This research sheds additional light on the underlying mechanisms of visual perception and perceptual learning and suggests that gamma oscillations and synchrony may be involved in the training-induced enhancement of figure-ground segregation.

Methods

Behavioral Experiments

The study and its experimental procedures were approved by the local Ethical Committee of the Faculty of Psychology and Neuroscience (ERCPN).

Participants

Eight healthy volunteers (6 female, mean age = 23.75, standard deviation = 6.453) participated in this study. Our study employed a repeated-measures design with extensive sampling, collecting a large number of trials from each participant. Sample size was determined based on comparable studies investigating visual perception and perceptual learning in humans (Intoy et al., 2024; Lange et al., 2020; Tesileanu et al., 2020). All participants had normal or corrected-to-normal visual acuity. After receiving full information about all procedures and the right to withdraw participation at any time, participants gave their written informed consent. All participants were compensated monetarily for their time.

Stimuli

Each texture stimulus consisted of a full-screen irregular grid of non-overlapping Gabor annuli with a diameter 0.7° , a spatial frequency of 5.7 cycles/degree and a mean luminance of 60.76 Cd/m^2 placed on a grey (60.76 Cd/m^2) background. Annuli contrasts were uniformly sampled from the full contrast range $U[0,1]$, except for a rectangular figure region $[(9 \pm 0.7)^\circ \times (5 \pm 0.4)^\circ]$ whose contrasts were drawn from a second uniform distribution $U[0.5 - \frac{\zeta}{2}, 0.5 + \frac{\zeta}{2}]$ with range ζ whose values were $\{0.01, 0.2575, 0.505, 0.7525, 1\}$. The figure region thus exhibited limited contrast heterogeneity, except when $\zeta = 1$ which is identical to the background (maximum) contrast heterogeneity. The center of the figure region was placed at an eccentricity of $(7 \pm 1)^\circ$. The polar angle of the figure was varied on each trial with the condition that it was always completely inside a single visual field quadrant. The coarseness of the grid was expressed as a factor ρ that scales the average center-to-center distance between any pair of neighboring annuli in the whole texture. The values of ρ were $\{1, 1.125, 1.250, 1.375, 1.5\}$. Each annulus was initially placed on a regular grid and subsequently slightly shifted in a random direction by a distance chosen from a uniform distribution that ranged from zero to half of the edge-to-edge distances of neighboring annuli. All combinations of ζ and ρ yield 25 unique stimulus conditions.

Tasks and Procedure

The experiment consisted of nine consecutive sessions (eight training and one transfer session) with a two-alternative forced choice design in which participants were required to indicate whether the rectangular figure was oriented horizontally or vertically by pressing the right and left arrow key, respectively. Responses were given with the middle and index fingers of the right hand. Each trial of the experiment started with the presentation of a fixation point (a small bright turquoise disk of $2^\circ \times 2^\circ$) for minimally 1000 ms, during which accurate fixation was to be initiated (i.e., deviation $< 2^\circ$ from fixation point) to trigger stimulus presentation. Participants were required to maintain fixation throughout presentation of the stimulus (1000 ms or less in case that a participant lost fixation or provided a response). Participants received feedback after each trial in the form of color changes (green correct; red incorrect) of the fixation point lasting for 500 ms. Feedback was followed by a 600 ms inter-trial interval during which an isoluminant (grey) screen was shown. When a participant's gaze fell outside the fixation window during the fixation period preceding the stimulus, or during stimulus presentation, the trial was aborted. Aborted trials were repeated at a randomly chosen time during the experiment.

The 25 conditions defined by contrast heterogeneity and grid coarseness were aggregated into experimental blocks such that all 25 combinations were shown exactly once per 25-trial block in random order. Each participant completed 30 blocks (750 trials) in each of the sessions. The figure

was placed in the lower right quadrant for the eight training sessions. In the transfer session, the figure was moved to the orthogonal (upper left) quadrant. Participants were made aware of the figure displacement but were not told in which quadrant to expect it.

The experiment was conducted in a dimly lit room. A chin and headrest were used to support the participant's head and to keep eye-screen distance constant at 57 cm. Stimuli were displayed on a 19" Samsung SyncMaster 940BF LCD monitor (Samsung, Seoul, South Korea; 60 Hz refresh rate, 1280 × 1024 resolution). Stimulus representation and response recording were performed by Psychtoolbox-3 for MATLAB 64-Bit (Version 3.0.14 - Build date: Apr 6th, 2018), under Microsoft Windows. Fixation was monitored with a desktop-mounted Eyelink 1000 eye-tracker (SR Research Ltd., 500 Hz or 1000 Hz sampling frequency, < 0.01° RMS spatial resolution, eye-movement data were down-sampled to 250 Hz).

Statistical Analyses

We used Bayesian hierarchical regression to analyze main and interactions of variables of interest which include manipulated stimulus features (contrast heterogeneity, grid coarseness and their interaction), model synchrony and learning effects (session and its interactions with stimulus features). Stimulus features and model synchrony were z-scored and session was mean-centered. Each of these statistical models included subjects-specific intercepts and slopes for all predictors. To investigate the relationship between the sizes of empirical and model Arnold tongues, we used Bayesian hierarchical regression with subject-level random intercepts. Because tongue size data entail only one measurement per subject per session, there was insufficient information to estimate subject-level slopes reliably.

All analyses were conducted using Bambi (v0.15.0) and ArviZ (v0.22.0) in Python. Priors were weakly informative defaults. Specifically, fixed effects were given Normal(0, 2.5), intercepts Normal(0, 2.5), and group-level standard deviations HalfNormal(2.5) priors. Correlations among random slopes and intercepts were given an LKJ(1) prior. Models were estimated using the No-U-Turn Sampler (NUTS) with 4 chains of 2,000 draws each, following a 2,000-draw tuning phase, for a total of 8,000 posterior samples. For all Bayesian models, convergence was assessed using standard diagnostics. All R^2 values were approximately 1.00, and all effective sample sizes (ESS) were sufficient, with the smallest ESS across analyses being 2200.

Oscillator Model of V1

We model a small patch of V1 that receives input from a $6.7^\circ \times 6.7^\circ$ square region of the visual field. The area of this square region matches the area of the rectangular figure region in our psychophysics experiments. The center of this region is furthermore located at an eccentricity matching that of the figure. We model this V1 patch as a network of weakly coupled phase oscillators arranged on an $n \times n$ ($n = 20$) irregular grid on the cortical surface. To that end we first defined a regular grid of receptive field centers for each oscillator in visual space and subsequently transformed receptive field coordinates to cortical coordinates of V1 using a complex-l(Balasubramanian & Schwartz, 2002; Schwartz, 1980; Schwartz, 2002; Schwartz, 1980) with generic human parameter values ($\alpha = 0.7$, $\alpha = 0.9$; Polimeni et al., 2005). While receptive fields are thus equally spaced in the visual field, neural oscillators themselves are not equally spaced on the cortical surface. The phase of each neural oscillator evolves according to a Kuramoto model:

$$\dot{\theta}_i = \omega_i + \frac{1}{N} \sum_{j=1}^N K_{ij}^s \sin(\theta_j - \theta_i), \quad i = 1, \dots, N; s = 1, \dots, 9 \quad (1)$$

where, θ_i is the phase of the i th oscillator, ω_i its intrinsic frequency, K_{ij}^s the coupling strength between oscillators i and j in session s (note that s is an index and not an exponent) and $N = n^2$ is the total number of oscillators. We treat the instantaneous frequency $\frac{\dot{\theta}}{2\pi}$ of each oscillator as a proxy for the instantaneous population firing rate of the corresponding neural assembly.

Intrinsic Frequency

In accordance with electrophysiological findings, the intrinsic frequency of each oscillator is a function of the local contrast in its receptive field (Roberts, Lowet, Brunet, TerWal, Tiesinga, Fries, & de Weerd, 2013). Specifically, the typical oscillation frequency ν (in Hz with corresponding $\omega = 2\pi\nu$) of a neural circuit in V1 is a linear function of local contrast (Lowet et al., 2015):

$$\nu = 25 + 0.25C \quad (2)$$

The local contrast received by each oscillator i is given by the weighted root-mean-squared (RMS) value of contrast (Frazor & Geisler, 2006):

$$C_i = \sqrt{\sum_{h=1}^M w_{ih} \frac{(L_h - \bar{L})^2}{\bar{L}^2}} / \sum_{h=1}^M w_{ih} \quad (3)$$

where L_h is the luminance of pixel h in the stimulus, $\bar{L}$ is the mean luminance over all pixels, and w_{ih} is the weight of pixel h and oscillator i . The weighting was specific to each oscillator as it reflects its unique receptive field which we modelled using an isotropic 2D Gaussian function:

$$w_{ih} = \exp\left(-\frac{(x_h - X_i)^2 + (y_h - Y_i)^2}{2\sigma_i^2}\right) \quad (4)$$

Here, (x_h, y_h) are the coordinates of the h^{th} pixel, while (X_i, Y_i) are the coordinates of the receptive field center of the i^{th} oscillator. In addition, σ_i is the size of the receptive field. We estimated receptive field sizes based on their location relative to the center of gaze. Specifically, receptive field diameter in V1 exhibits a threshold linear relationship with receptive field eccentricity (e ; Freeman & Simoncelli, 2011) such that $\phi = \max(0.172e - 0.25, 1)$. We related the receptive field diameter to the standard deviation of a Gaussian in two steps. First, we related the diameter to the full width at half maximum (FWHM) of a Gaussian beam $FWHM = \frac{\sqrt{\ln 2}}{\sqrt{2}} \phi$ (Hill, 2007). Then, we related the FWHM to the standard deviation $\sigma = \frac{FWHM}{2\sqrt{\ln 2}}$. Combining these steps, the standard deviation is one fourth of the receptive field diameter.

Adaptive Coupling

The coupling strength K_{ij}^1 between pairs of oscillators in the first session is a function of their cortical distance:

$$K_{ij}^1 = \gamma e^{-\lambda d_{ij}} \quad (5)$$

Here, γ is the maximum coupling strength and λ controls how fast the coupling strength decreases as a function of cortical distance (d_{ij}) between oscillators i and j . We estimated γ and λ from previously published data relating coupling strength to cortical distance within V1 in two macaque monkeys (Lowet, Roberts, Peter, Gips, & de Weerd, 2017).

Coupling strength in the remaining sessions is the result of an offline learning process that takes the experience accumulated over an individual training session into account. Specifically, learning in our model depends on the pairwise phase-locking value (PLV; Lachaux et al., 1999) between model oscillators accumulated over trials within one session. Phase-locking values were computed over the second half of the simulation period, which was subsampled to 50 timepoints.

Accumulation across trials involves summing PLVs over trials, where the contribution of each trial is weighted by the probability that the model would produce a correct response on that trial. The weighted PLV is summarized in a matrix Q . To obtain the probability of a correct response (P_c) from model simulations, we related it to the degree of synchrony (r) among phase oscillators through a psychometric function:

$$P_c = 1/[1 + \exp(-\mu_0 - \mu_1 r)] \quad (6)$$

Parameters of this function (i.e., μ_0 and μ_1) were estimated based on model simulations and empirical results from the first session. The temporal evolution of pairwise coupling strength is given by a Hebbian-type learning rule:

$$\dot{K}_{ij} = \epsilon(\gamma Q_{ij}^s - K_{ij}). \quad (7)$$

Here, ϵ is a learning rate. Essentially, pairwise structural coupling approaches pairwise functional coupling, as measured by the weighted PLV within a session (Q^s), scaled by the maximum coupling strength γ . Integration of Equation 5 with respect to time yields

$$K_{ij}^{s+1} = \exp(-\epsilon t) K_{ij}^s + [1 - \exp(-\epsilon t)] \gamma Q_{ij}^s. \quad (8)$$

Here, t is the time between two sessions during which learning occurs (e.g., during sleep). Since neither t nor ϵ can be measured independently and are not known a priori, we merged them into a single free parameter $E = \epsilon t$. We refer to this as the effective learning rate. We adjusted the parameter E to maximize the correspondence, measured by the weighted Jaccard similarity, between the distribution of performance observed in the *second* experimental session and the distribution of synchrony after letting the model learn according to Equation 6. To that end we used a coarse-to-fine grid search wherein we let the model learn using a grid of 25 candidate effective learning rates and selected the value that enabled best prediction of session 2 performance. We then created a new, finer, grid around the best effective learning rate and repeated this procedure. In total, we explored five nested grids. Note that the learning procedure depends on data from the first two sessions to establish a mapping from synchrony to performance (parameters μ_0 and μ_1 of the psychometric function linking synchrony to performance) and to estimate the effective learning rate, respectively. We kept these parameters fixed for predicting the results of sessions 3-8. To further disentangle data used for parameter tuning and data used for testing model predictions, we utilized a leave-one-out cross-validation procedure. We estimated all parameters from the first two sessions of seven of our eight participants, and then predicted results of session 3-8 in the left-out participant. We repeated this procedure eight times, once per participant, and stored all results for further analysis.

Simulations

We simulated eight training sessions, each consisting of 30 blocks with 25 trials. Within each trial, we simulated a one-second stimulus monitoring interval assigned to a specific combination of contrast heterogeneity and grid coarseness. All simulations were performed in Python 3.12.2 using the `odeint` method from `scipy`'s (version 1.12.0) `integrate` submodule. For each simulated trial, we evaluated synchrony by measuring the radius ($r \in [0,1]$) of the Kuramoto order parameter given by

$$r e^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j} \quad (9)$$

where θ_j is the phase of the oscillator j . For each simulated trial, r was averaged within the second half of the trial duration, and over all blocks.

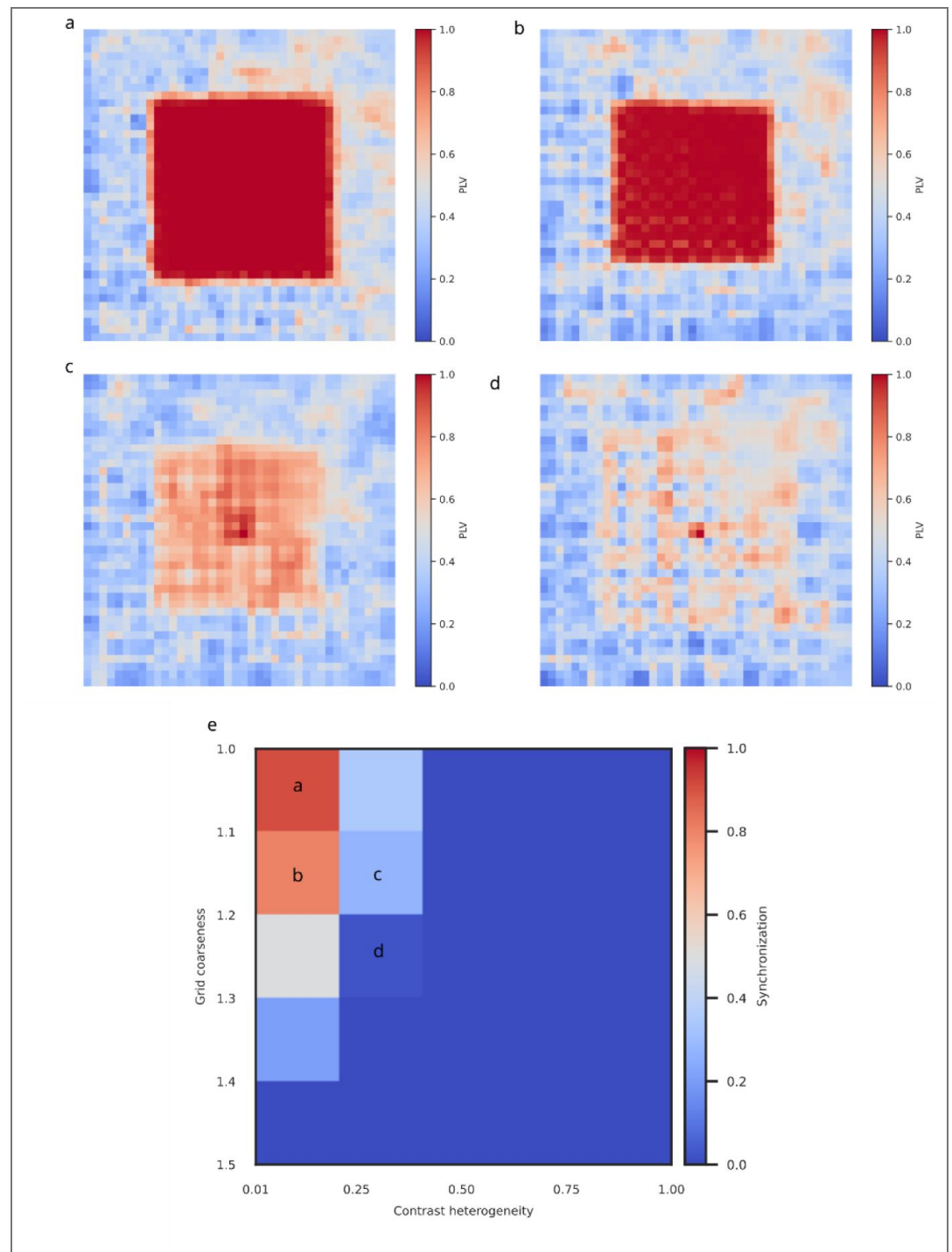
System Specifications

All analyses and simulations were performed as a Docker containerized Snakemake workflow executed on a single compute node of Maastricht University's Data Science Research Infrastructure (DSRI). The node is equipped with two AMD EPYC 7551 32-Core Processors, has a nominal 512 GB of RAM, and operates on Fedora 37. The workflow utilized 30 of 64 available cores to simulate all blocks of a particular trial in parallel. To ensure that all results can be reproduced exactly, the random seed of our workflow was fixed at 1709026616.

Supplementary Materials

Validation of the Figure-Only Model

To validate our figure-only modeling approach, we conducted simulations that included both the figure and a surrounding high-heterogeneity background. We visualized the emergence of a synchronized assembly by calculating the Phase-Locking Value (PLV) of every oscillator relative to a reference oscillator at the center of the figure.



Supplementary Figure 1. **a-d**, Phase-locking values of every oscillator relative to a reference oscillator positioned at the center of the figure for four different stimulus conditions. Phase-locking values are averages over 20 simulations. **e**, The Arnold Tongue from our figure-only simulations reference, with labels indicating the four representative conditions selected for the full simulation.

Design analysis

With a sample size of 8 participants, the detection probability (power) exceed 90% for all effects and robustness to both type-S and type-M errors.

Sample size	Detection probability			Type-S error			Type-M error		
	Contrast Heterogeneity	Grid Coarseness	Interaction	Contrast Heterogeneity	Grid Coarseness	Interaction	Contrast Heterogeneity	Grid Coarseness	Interaction
4	0.74	0.54	0.48	0.00	0.00	0.00	0.94	1.10	1.11
6	0.92	0.82	0.84	0.00	0.00	0.00	1.11	1.01	1.03
8	0.96	0.92	0.92	0.00	0.00	0.00	0.99	0.96	1.02

Table 2. Design analysis of main analysis in session 1.

Detection probability refers to the proportion of simulated datasets in which the posterior probability of an effect exceeded 0.95 in the predicted direction. Type-S error indicates the probability of detecting an effect, but in the wrong direction (sign reversed). Type-M error refers to the ratio of estimated to true effect size when detected. A value of 1 indicates no deviation whereas values larger (smaller) than 1 indicate that effects are over (under) estimated.

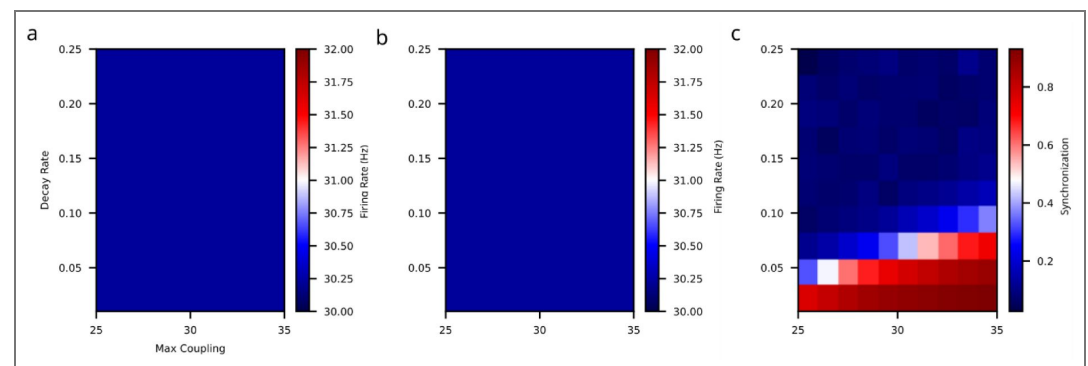
Average Firing is Insensitive to Oscillator Interactions

We demonstrate that average feedforward (intrinsic) firing rates and average effective firing rates emerging from oscillator interactions within our model are identical. We first show that average intrinsic and effective firing rates are identical across all 25 experimental conditions, using the fixed coupling parameters ($\gamma=24.63$, $\lambda=0.22$) used in the main text. For each of the 25 conditions, we computed the mean intrinsic and mean effective firing rates across all oscillators; i.e., across the entire figure region. The total absolute difference between average firing rates summed over all 25 conditions was negligible ($< 10^{-12}$ Hz).

To demonstrate that this is a general property of the model and not specific to our chosen parameter set, we performed a second analysis. We simulated a single, representative stimulus condition (contrast heterogeneity = 0.5, grid coarseness = 1.5) across a range of global coupling parameters (10 levels of maximum coupling strength γ and 10 levels of coupling decay λ).

For each of the resulting 100 simulations, we calculated the average intrinsic and effective firing rates across all oscillators. The average effective firing rate remained identical to the average intrinsic firing rate across this entire parameter space. The total absolute difference summed over all parameter combinations was again negligible ($< 10^{-11}$ Hz). This result is visualized in Figure S1. While synchrony varies across the parameter space, average firing rate remains constant and is determined exclusively by the stimulus.

Note that this does not imply that firing rates of individual oscillators are not affected by interactions with other oscillators. The fact that they can synchronize for certain stimulus conditions and/or parameter combinations shows that individual firing rates change, but they vary consistently around a fixed mean. Interactions in the model reduce the variance of firing rates (i.e., produce synchrony).



Supplementary Figure 2. Model derived quantities for different combinations of maximum coupling and decay rate. a, intrinsic firing rate averaged over all oscillators. **b,** effective firing rate averaged over all oscillators. **c,** In-phase synchronization among all oscillators. Neither average intrinsic nor average effective firing rates are sensitive to model parameters and are both fixed at 30.34 Hz.

Data availability

All data generated or analyzed during this study are openly accessible at <https://zenodo.org/doi/10.5281/zenodo.10817186>. The code for data acquisition can be accessed at <https://github.com/ccnmaastricht/TextureStimuli-FigureGround.git>. The code for performing all analyses and simulations can be accessed at <https://github.com/ccnmaastricht/NeuralSynchrony-FigureGround>.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This paper by Karimian et al proposes an oscillator model tuned implementing binding by (gamma) synchrony principles in a visual task. The authors set out to show how well these principles explain human behavior in a figure-ground segregation tasks. The model is inspired by electrophysiological findings in non-human primates suggesting that gamma oscillations in early visual cortex implement feature-binding through a synchronization of feature-selective neurons. The psychophysics experiment involves the identification of a

figure consisting of gabor annuli, presented on a background of gabor annuli. The participants' task is to identify the orientation of the figure. The task difficulty is varied based on the contrast and density of the gabor annuli that make up the figure. The same figures are used as inputs to the oscillator model. The authors report that both the discrimination accuracy in the psychophysics experiment and the synchrony of the oscillators in the proposed model follow a similar "Arnold Tongue" relationship when depicted as a function of the texture-defining features of the figure. This finding is interpreted as evidence for gamma synchrony being the underlying mechanism of the figure-ground segregation.

Strengths:

The design of the proposed model is well-informed by electrophysiological findings, and the idea of using computational modeling to bridge between intracranial recordings in non-human primates and behavioral results in human participants is interesting. Previous work has criticized the gamma synchrony theories based on the observation that synchronization in the gamma-band is highly localized and the frequency of the oscillation depends on the visual features of the stimulus. I appreciate how the authors demonstrate that frequency-dependence and local synchronization can be features of gamma synchrony, and not contradictory to the theory. As such, I feel that this work has the potential to contribute meaningfully to the debate on whether binding by gamma synchrony is a biophysically realistic model of feature-binding in visual cortex.

I also acknowledge the additional simulations the authors present in this version of the manuscript, showing that the model is able to segregate figure from ground.

Weaknesses:

The authors have addressed my previous concerns regarding the quantification of effect sizes. I also appreciate the authors argument that the results support the idea of feature-binding through synchronization in the gamma-band, as the model's parameters were informed by electrophysiological recordings from non-human primates. Personally, I would have been curious to see if the intrinsic frequencies of the model are indeed in the gamma-band, I don't believe the authors include a figure on that. Weaknesses are still the absence of electrophysiological recordings to support the frequency-specificity of the claims, e.g. in the form of EEG/MEG recordings, but I understand that these may be difficult to obtain, as gamma oscillations are relatively weak in response to static gratings. As the authors emphasize in this updated version, they present one possible mechanism of feature binding that is not contrasted to alternative mechanisms such as binding by increased firing rates. Understandably, implementing a second model would be out of scope.

The presented simulations and behavioural results support the authors aim of presenting an oscillator model informed by gamma synchronization in V1 that supports figure-ground segregation.

Likely impact:

This work makes several predictions about the degree of synchronization for different visual properties of the figure, that could be tested with electrophysiological methods. I therefore believe that the paper has the potential to motivate interesting follow-up studies to understand how visual cortex solves the binding problem.

Comment on revised version:

In this reviewed version of the manuscript, the authors present several follow-up simulations and clarifications that address previously outlined weaknesses.

<https://doi.org/10.7554/eLife.105482.2.sa2>

Reviewer #2 (Public review):

The authors aimed to investigate whether gamma synchrony serves a functional role in figure-ground perception. They specifically sought to test whether the stimulus-dependence of gamma synchrony, often considered a limitation, actually facilitates perceptual grouping. Using the theory of weakly coupled oscillators (TWCO), they developed a framework wherein synchronization depends on both frequency detuning (related to contrast heterogeneity) and coupling strength (related to proximity between visual elements). Through psychophysical experiments with texture discrimination tasks and computational modeling, they tested whether human performance follows patterns predicted by TWCO and whether perceptual learning enhances synchrony-based grouping.

Strengths:

- (1) The theoretical framework connecting TWCO to visual perception is innovative and well-articulated, providing a potential mechanistic explanation for how gamma synchrony might contribute to both feature binding and separation.
- (2) The methodology combines psychophysical measurements with computational modeling, with a solid quantitative agreement between model predictions and human performance.
- (3) In particular, the demonstration that coupling strengths can be modified through experience is remarkable and suggests gamma synchrony could be an adaptable mechanism that improves with visual learning.
- (4) The cross-validation approach, wherein model parameters derived from macaque neurophysiology successfully predict human performance, strengthens the biological plausibility of the framework.

Likely Impact and Utility:

This work offers a fresh perspective on the functional role of gamma oscillations in visual perception. The integration of TWCO with perceptual learning provides a novel theoretical framework that could influence future research on neural synchrony.

The computational model, with parameters derived from neurophysiological data, offers a useful tool for predicting perceptual performance based on synchronization principles. This approach might be extended to study other perceptual phenomena and could inspire designs for artificial vision systems.

The learning component of the study may have a particular impact, as it suggests a mechanism by which perceptual expertise develops through modified coupling between neural assemblies. This could influence thinking about perceptual learning more broadly, but also raises questions about the underlying mechanism.

Additional Context:

Historically, the functional significance of gamma oscillations has been debated, with early theories of temporal binding giving way to skepticism based on gamma's stimulus-dependence. This study reframes this debate by suggesting that stimulus-dependence is exactly what makes gamma useful for perceptual grouping.

The successful combination of computational neuroscience and psychophysics is a significant strength of this study.

The field would benefit from future work extending (if possible) these findings to more naturalistic stimuli and directly measuring neural activity during perceptual tasks.

Additionally, studies comparing predictions from synchrony-based models against alternative mechanisms would help establish the specificity of the proposed framework.

Comments on revised version:

The authors now soften their claim. However, the paper demonstrates that TWCO-derived predictions quantitatively match human figure-ground perception in texture stimuli, and that a synchrony-based readout provides a viable mapping from stimulus to behavior. Given that they cite (and do not show in this paper) the link to synchrony, what they actually establish is that this particular transformation of stimulus features maps better onto behavior. That's meaningful, but it is not a demonstration of mechanism.

<https://doi.org/10.7554/eLife.105482.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper by Karimian et al proposes an oscillator model tuned to implement binding by synchrony (BBS) principles in a visual task. The authors set out to show how well these BBS principles explain human behavior in figure-ground segregation tasks. The model is inspired by electrophysiological findings in non-human primates, suggesting that gamma oscillations in early visual cortex implement feature-binding through a synchronization of feature-selective neurons. The psychophysics experiment involves the identification of a figure consisting of gabor annuli, presented on a background of gabor annuli. The participants' task is to identify the orientation of the figure. The task difficulty is varied based on the contrast and density of the gabor annuli that make up the figure. The same figures (without the background) are used as inputs to the oscillator model. The authors report that both the discrimination accuracy in the psychophysics experiment and the synchrony of the oscillators in the proposed model follow a similar "Arnold Tongue" relationship when depicted as a function of the texture-defining features of the figure. This finding is interpreted as evidence for BBS/gamma synchrony being the underlying mechanism of the figure-ground segregation.*

Note that I chose to use "BBS" over gamma synchrony (used by the authors) in this review, as I am not convinced that the authors show evidence for synchronization in the gamma-band.

We thank the reviewer for their careful assessment of our manuscript and useful comments that we believe have served to strengthen our work.

Strengths:

The design of the proposed model is well-informed by electrophysiological findings, and the idea of using computational modeling to bridge between intracranial recordings in non-human primates and behavioral results in human participants is interesting. Previous work has criticized the BBS synchrony theory based on the observation that synchronization in the gamma-band is highly localized and the frequency of the oscillation depends on the visual features of the stimulus. I appreciate how the authors demonstrate that frequency-dependence and local synchronization can be features of BBS, and not contradictory to the theory. As such, I feel that this work has the potential to

contribute meaningfully to the debate on whether BBS is a biophysically realistic model of feature-binding in visual cortex.

Weaknesses:

I have several concerns regarding the presented claims, assessment of meaning and size of the presented effects, particularly with regard to the absence of a priori defined effect sizes.

Firstly, the paper makes strong claims about the frequency-specificity (i.e., gamma synchrony) and anatomical correlates (early visual cortex) of the observed effects. These claims are informed by previous electrophysiological work in non-human primates but are not directly supported by the paper itself. For instance, the title contains the word "gamma synchrony", but the authors do not demonstrate any EEG/MEG or intracranial data in from their human subjects supporting such claims, nor do they demonstrate that the frequencies in the oscillator model are within the gamma band. I think that the paper should more clearly distinguish between statements that are directly supported by the paper (such as: "an oscillator model based on BBS principles accounts for variance in human behavior") and abstract inferences based on the literature (such as "these effects could be attributed to gamma oscillations in early visual cortex, as the model was designed based on those principles").

We thank the reviewer for this helpful comment and agree that the scope of our claims should be clearly delineated between what is directly supported by our data and what is theoretically inferred from prior literature.

We revised the Abstract, Introduction, and early Discussion to moderate the strength of our statements and make the distinction explicit. The revised title now emphasizes that our study tests principles derived from prior work on gamma synchrony rather than directly demonstrating gamma activity in humans. Throughout the text, we use more cautious phrasing that highlights potential mechanisms and theoretical predictions. The intention of our study was not to position synchrony as the only viable mechanism of figure-ground perception. Rather, our goal was to reinvigorate it as a potential contender by showing that features often cited as limitations of synchrony-based binding may in fact be essential properties of the mechanism. We updated phrasing throughout the manuscript to make this clearer and avoid overstating the study's contribution.

Importantly, our model is not agnostic with respect to frequency band. Oscillator frequencies exhibited by model units are within the gamma range by design. Frequency emerges directly from the contrast within each oscillator's receptive field, following an empirically established relationship between stimulus contrast and gamma frequency. To our knowledge, such a robust, quantitative relationship between stimulus features to exact oscillation frequency has not been consistently demonstrated for other frequency bands. This relationship yields gamma-band frequencies for all contrasts used in our simulations. The model is thus indeed a gamma oscillator model of V1, not a generic instantiation of Binding by Synchrony (BBS) principles.

That said, we fully agree with the reviewer that our study cannot demonstrate a direct link between gamma synchrony in visual cortex and human behavior. Our behavioral and modeling results instead show that synchronization principles derived from gamma-band physiology in V1 can predict perceptual performance patterns. We now make this distinction explicit throughout the revised manuscript.

Secondly, unlike the human participants, the model strictly does not perform figure-ground segregation, as it only receives the figure as an input.

We thank the reviewer for the opportunity to clarify our modeling approach. We chose not to model the background to reduce computational cost, since including it requires a substantially larger number of oscillators without changing the model's predictions. The model thus indeed only receives the figure region as input. We aimed to test the local grouping mechanism predicted by TWCO, rather than to simulate a full figure-ground segregation process including a read-out stage. Our model therefore isolates the conditions under which local synchrony emerges within the figure region, assuming that a downstream read-out mechanism (not explicitly modeled here) would detect regions of coherent activity. The exact nature of such a read-out mechanism was beyond the scope of our work.

To confirm that our simplified model is a valid proxy, we ran additional simulations including the background and found that a coherent figure assembly reliably emerges, as can be seen in the phase-locking patterns relative to a reference oscillator at the center of the figure. This validates that the principles of local grouping we studied in isolation hold even when the figure is embedded in a noisy surround. We have added an explicit note in the Results (paragraph 2) that we only simulate the figure and added Supplementary Figure S1 showing the additional simulations.

Finally, it is unclear what effect sizes the authors would have expected a priori, making it difficult to assess whether their oscillator model represents the data well or poorly. I consider this a major concern, as the relationship between the synchrony of the oscillatory model and the performance of the human participants is confounded by the visual features of the figure. Specifically, the authors use the BBS literature to motivate the hypothesis that perception of the texture-defined figure is related to the density and contrast heterogeneity of the texture elements (gabor annuli) of the figure. This hypothesis has to be true regardless of synchrony, as the figure will be easier to spot if it consists of a higher number of high-contrast gabors than the background. As the frequency and phase of the oscillators and coupling strength between oscillators in the grid change as a function of these visual features, I wonder how much of the correlation between model synchrony and human performance is mediated by the features of the figure. To interpret to what extent the similarity between model and human behavior relies on the oscillatory nature of the model, the authors should find a way to estimate an empirical threshold that accounts for these confounding effects. Alternatively, it would be interesting to understand whether a model based on competing theories (e.g., Binding by Enhanced Firing, Roelfsema, 2023) would perform better or worse at explaining the data.

We thank the reviewer for these insightful and constructive comments, which have prompted additional analyses that we believe substantially strengthen our work. The reviewer raises two main points: (1) the need for a benchmark to assess our model's performance, and (2) the concern that the relationship between model synchrony and behavior might be a non-causal "confound" of the visual features. We address each point below.

(1) Benchmarking model performance

We agree that it is important to assess how well our model performs relative to the data and included this in the original manuscript. We did not predefine an absolute good fit threshold because absolute agreement depends on irreducible noise and inter-subject variability, making a universal cutoff arbitrary. Instead, we had benchmarked model performance in two complementary ways. First, the noise ceiling shown in Figure 5 provides an empirical benchmark for the maximum fit any model could achieve on our data. Simulated Arnold tongues (based on synchrony) approach this ceiling achieving 89% of possible similarity for correlation and 79% of possible similarity for weighted Jaccard similarity, respectively. Second, the parameter sweep (Figure 3) situates our model's performance within the broader parameter space. It shows that the model, whose key parameters were fixed a priori from

independent macaque neurophysiological data, lies close to the optimal regime for explaining the human data. It also provides an estimate of the lower bound (worst-performing point) on the fit that a misspecified model implementing the identical mechanism would achieve. Our model with fixed a priori parameters does 1.41 times better than a misspecified model for the correlation fit metric and 3 times better for weighted Jaccard similarity.

(2) Synchrony as mechanism vs. potential confound

We appreciate the reviewer's suggestion to test whether synchrony explains behavior beyond stimulus features. In our framework, synchrony is a near-deterministic function of the manipulated stimulus features given fixed model parameters. As a result, synchrony and the stimulus features are collinear ($R^2 \approx 0.8$) leaving no independent variance for synchrony to explain once stimulus features are included. Adding both into one statistical model yields unstable coefficients and no out-of-sample improvement.

Mechanistically, we believe the relevant question is not whether synchrony explains behavior beyond stimulus features but whether synchrony is the correct transformation of the stimulus features to reproduce the behavioral pattern. Please note that in our design we ensured that mean contrast and luminance are identical in the figure and the background such that there are not more high-contrast Gabors in the figure than in the background. We did this with the aim to render mean contrast not a relevant feature. However, there are more high-contrast Gabors in the background, and it is conceivable that the absence of such high contrasts in the figure drives the detection/discrimination of the figure. We therefore agree that testing alternative models would further clarify the unique explanatory value of the synchrony mechanism. To that end, we derived two alternative rate-based readouts from the same V1 simulations of our model from which we derived synchrony. First, average firing rates inside the figure and second, the difference between average firing rates inside the figure and average firing rates in the background (rate difference). We analyzed each individually as predictors of behavior and performed a model comparison based on out-of-sample predictions. While rate difference (but not average firing) showed meaningful associations with performance when considered alone, the synchrony readout had a larger effect size and was favored by the model comparison. We added a new subsection comparing synchrony to rate-based alternatives in the Results (paragraphs 7-9), including additional Bayesian analyses and LOO-CV model comparison. Please note that the model comparison we added to the manuscript provides an additional benchmark beyond the map-level ceiling analysis. It indicates that the mapping from stimulus features to behavior via synchrony generalizes best without requiring an a priori good-fit threshold.

We agree that formally comparing our model to a sophisticated rate-based alternative, such as an instantiation of the Binding by Enhanced Firing model, is an important direction for future work. However, it remains an open and non-trivial question whether such a model could quantitatively reproduce the precise shape of the behavioral Arnold tongue that emerges from the systematic manipulation of our stimulus parameters. Implementing and parameterizing such a model in a comparable, biologically grounded framework is a substantial undertaking that lies beyond the scope of the current study. Therefore, our goal here was not to claim exclusivity for synchrony-based mechanisms, but rather to re-evaluate their plausibility by showing that features often seen as limitations (stimulus dependence and frequency heterogeneity) are, in fact, essential characteristics of the TWCO framework that can predict complex behavioral outcomes.

We would also like to clarify that our stimulus features were derived from theory rather than psychophysical literature. Starting from the principles of TWCO, we mapped frequency detuning and coupling strength onto known anatomical and physiological properties of early visual cortex, and only then derived the corresponding stimulus manipulations (contrast heterogeneity and grid coarseness). Demonstrating that these features predict behavior is

therefore not trivial but constitutes a first empirical confirmation that the core TWCO variables match perception.

Apart from adding analyses of additional rate-based readouts of our model, we also refined our discussion of the relationship between these and a synchrony-based mechanism.

Reviewer #2 (Public review):

The authors aimed to investigate whether gamma synchrony serves a functional role in figure-ground perception. They specifically sought to test whether the stimulus-dependence of gamma synchrony, often considered a limitation, actually facilitates perceptual grouping. Using the theory of weakly coupled oscillators (TWCO), they developed a framework wherein synchronization depends on both frequency detuning (related to contrast heterogeneity) and coupling strength (related to proximity between visual elements). Through psychophysical experiments with texture discrimination tasks and computational modeling, they tested whether human performance follows patterns predicted by TWCO and whether perceptual learning enhances synchrony-based grouping.

We thank the reviewer for their thoughtful and constructive review. We believe the comments have served to improve our work.

Strengths:

(1) The theoretical framework connecting TWCO to visual perception is innovative and well-articulated, providing a potential mechanistic explanation for how gamma synchrony might contribute to both feature binding and separation.

(2) The methodology combines psychophysical measurements with computational modeling, with a solid quantitative agreement between model predictions and human performance.

(3) In particular, the demonstration that coupling strengths can be modified through experience is remarkable and suggests gamma synchrony could be an adaptable mechanism that improves with visual learning.

(4) The cross-validation approach, wherein model parameters derived from macaque neurophysiology successfully predict human performance, strengthens the biological plausibility of the framework.

Weaknesses:

(1) The highly controlled stimuli are far removed from natural scenes, raising questions about generalisability. But, of course, control (almost) excludes ecological validity. The study does not address the challenges of natural vision or leverage the rich statistical structure afforded by natural scenes.

We agree with the reviewer that the insights of the present study are limited to texture stimuli and have made adjustments in the Discussion (final two paragraphs) to avoid claiming generalizability to natural stimuli. We have also adjusted the title to specifically limit our results to texture stimuli. To establish the principles of TWCO, we needed tight control over the stimulus, but are intrigued by the idea to investigate natural scenes. We have added to our Discussion (paragraph 9) that future should evaluate to what extent the principles we investigate here apply to natural scenes. Synchrony-based mechanisms have been successfully used for image segmentation tasks in machine vision, showing that the proposed mechanism can in principle work for natural scenes.

(2) The experimental design appears primarily confirmatory rather than attempting to challenge the TWCO framework or test boundary conditions where it might fail.

We thank the reviewer for this important point. Our primary motivation was to address the neurophysiological properties of gamma synchrony that have been suggested to severely challenge the binding by synchrony mechanism. Particularly the strong dependence of gamma oscillations and synchrony on stimulus features. Our goal was to show that from the perspective of TWCO, these challenges become expected components of the mechanism. In essence, we wanted to promote a conceptual shift that converts what pushes a theory to its limit into something that is actually its central tenet. To facilitate this shift, we designed the experiment to directly test this core tenet.

While our approach was designed to test a central prediction of TWCO rather than explicitly challenge its boundaries, we respectfully argue that it was far from a simple confirmatory experiment. The design incorporated high-risk elements that provided considerable room for both the theory and our model to fail. First, the core prediction itself was non-obvious and highly specific. We did not simply test whether contrast heterogeneity and grid coarseness affect perception. We tested the stronger hypothesis that they would reflect a specific, interactive trade-off (the behavioral Arnold tongue) as specified by TWCO. Second, our modeling approach was deliberately constrained to provide a further stringent test. We did not post-hoc optimize the model's key parameters to fit our behavioral data. Instead, we fixed them a priori based on independent neurophysiological data from macaques. This was a high-risk choice, as a mismatch between a priori model predictions and the human data would have seriously challenged the framework's generalizability.

We agree that future research should further challenge TWCO. For instance, by using stimuli that require segregating several objects simultaneously or objects that cover more extensive regions of the visual field.

(3) Alternative explanations for the observed behavioral effects are not thoroughly explored. While the model provides a good fit to the data, this does not conclusively prove that gamma synchrony is the actual mechanism underlying the observed effects.

We agree that our results do not conclusively show that gamma synchrony is the actual mechanism underlying figure-ground segregation. We admit that the original phrasing used throughout the manuscript was too strong and gave the impression that we wanted to establish exactly that. However, the goal of our work was only to reinvigorate gamma synchrony as a potential contender by showing that features often cited as limitations of synchrony-based binding may in fact be essential properties of the mechanism. We have revised the title and made adjustments throughout the manuscript to better reflect this more moderate goal.

Additionally, we added tests of alternatives (Results, paragraphs 7–9) to clarify the unique explanatory value of the synchrony mechanism. To that end, we derived two alternative rate-based readouts from the same V1 simulations of our model. First, we extracted average firing rates inside the figure. Second, we computed the difference between average firing rates inside the figure and average firing rates in the background (rate difference). We analyzed each individually as predictors of behavior and performed a model comparison between these two and synchrony based on out-of-sample predictions. While the rate difference (but not average firing) showed meaningful associations with performance when considered alone, the synchrony readout had a larger effect size and was favored by the model comparison.

(4) Direct neurophysiological evidence linking the observed behavioral effects to gamma synchrony in humans is absent, creating a gap between the model and the neural mechanism.

We agree that the model only provides a how-possibly account linking stimulus features to performance. Showing that the brain actually relies on this mechanism would require showing that cortical synchrony mediates the effect of stimulus features on behavior beyond firing rates. Collecting such data would constitute a major effort that would go beyond the scope of this study. We acknowledge the need for electrophysiological data and the mediation analysis in the updated Discussion.

Achievement of Aims and Support for Conclusions:

The authors largely achieved their primary aim of demonstrating that human figure-ground perception follows patterns predicted by TWCO principles. Their psychophysical results reveal a behavioral "Arnold tongue" that matches the synchronization patterns predicted by their model, and their learning experiment shows that perceptual improvements correlate with predicted increases in synchrony.

The evidence supports their conclusion that gamma synchrony could serve as a viable neural grouping mechanism for figure-ground segregation. However, the conclusion that "stimulus-dependence of gamma synchrony is adaptable to the statistics of visual experiences" is only partially supported, as the study uses highly controlled artificial stimuli rather than naturalistic visual statistics, or shows a sensitivity to the structure of experience.

Likely Impact and Utility:

This work offers a fresh perspective on the functional role of gamma oscillations in visual perception. The integration of TWCO with perceptual learning provides a novel theoretical framework that could influence future research on neural synchrony.

The computational model, with parameters derived from neurophysiological data, offers a useful tool for predicting perceptual performance based on synchronization principles. This approach might be extended to study other perceptual phenomena and could inspire designs for artificial vision systems.

The learning component of the study may have a particular impact, as it suggests a mechanism by which perceptual expertise develops through modified coupling between neural assemblies. This could influence thinking about perceptual learning more broadly, but also raises questions about the underlying mechanism that the paper does not address.

Additional Context:

Historically, the functional significance of gamma oscillations has been debated, with early theories of temporal binding giving way to skepticism based on gamma's stimulus-dependence. This study reframes this debate by suggesting that stimulus-dependence is exactly what makes gamma useful for perceptual grouping.

The successful combination of computational neuroscience and psychophysics is a significant strength of this study.

The field would benefit from future work extending (if possible) these findings to more naturalistic stimuli and directly measuring neural activity during perceptual tasks. Additionally, studies comparing predictions from synchrony-based models against alternative mechanisms would help establish the specificity of the proposed framework.

Recommendations for the authors:

Reviewing Editor Comments:

In a joint discussion to integrate the peer reviews and agree on the eLife recommendations, both reviewers agreed that the work is valuable, but they were on the fence about whether the strength of evidence was incomplete or solid, eventually settling on incomplete. The reviewers make several recommendations for improving these ratings, which I (Reviewing Editor) have organised into 3 points below, with point 1 of particular importance. Underneath the summary, please see the individual recommendations of the reviewers.

(1) Strengthen evidence for the unique role of gamma synchrony in explaining the data, and ensuring claims are directly supported by relevant data:

Reviewers 2 and 3 both note the lack of direct evidence for gamma involvement, and reviewer 2 observes that the fit with behaviour may trivially be explained by a relationship between contrast heterogeneity and grid coarseness without need for oscillation. The reviewers felt that the approach of fitting the model to human data could be strengthened to help address this issue - and they offer various solutions, e.g., more principled a-priori criteria around good vs bad fit of the model to both main task and training data, and comparison to alternative binding models (Reviewer 2), identifying and testing boundary conditions of the model (Reviewer 3). There is also the possibility of collecting direct human neurophysiological evidence linking the behavioural data to neural mechanisms. Our discussion also highlighted the need to weaken claims (including in the title) where links are not directly demonstrated by methods from the present study, e.g., resting on indirect comparisons to primate literature.

We agree with the editor and reviewers that this was a critical point. To address it, we have made several major revisions.

As suggested, we have weakened claims where the links are not directly demonstrated by our data. The title has been revised to be more specific, and we have carefully edited the abstract, introduction, and discussion to distinguish between our model's predictions and direct neurophysiological evidence.

To address the concern that our model's fit might be trivially explained by visual features, we have performed a new analysis comparing the synchrony-based readout to two alternative rate-based readouts from the same V1 simulations. This new comparison shows that the synchrony readout provides a superior out-of-sample prediction of human behavior.

While a full implementation of a competing theory like "Binding by Enhanced Firing" would be a valuable next step, we note that parameterizing such a model in a comparably grounded framework is a substantial undertaking beyond the scope of the present study. Our new analysis provides an important first step in this direction.

(2) Make explicit and address the limitations of the stimuli:

Include that the model is not extracting the figure from the background, and the controlled stimuli may limit generalizability.

To address the concern that our model was not performing true figure-ground extraction, we performed a new set of simulations that included both the figure and the immediate background. The results confirm that synchrony dynamics within the figure region are not affected by the presence of the background. We added these validation results as supplementary materials. We have additionally made the modeling choice and its justification more explicit in the Results and Methods sections.

We have revised the Discussion to be more explicit about the limitations of using highly controlled texture stimuli. We now clearly state that our findings are specific to this context and that further research is required to determine if these principles generalize to the segregation of objects in natural scenes.

(3) Some clarifications to make more accessible:

Include the figure explaining the framework (Reviewers 1&2), and also the model details (Reviewer 2).

We have revised Figure 1 and its caption to more clearly illustrate the links from TWCO principles to their neural implementation in V1 and the resulting behavioral predictions.

We have expanded the Methods section to provide a more detailed and accessible description of the model's construction. We now clarify precisely how the oscillator grid was defined in visual space, how eccentricity-dependent receptive field sizes were implemented, and how these were mapped onto a retinotopic cortical surface to determine coupling strengths.

Reviewer #1 (Recommendations for the authors):

(A) Major concerns:

(1) My main concern:

My main concern is the repeated claims that the observed findings can be attributed to gamma synchrony in the early visual cortex. I find this claim misleading as the authors do not report any electrophysiological data that directly supports such claims. As stated in my public review, I feel that the authors should be clear about direct evidence versus more abstract inferences based on the literature.

In particular, I recommend changing claims about "gamma synchrony" to "Binding by Synchrony" That being said, the authors can outline that the model was built under the assumption that this synchrony is mediated by gamma in early visual cortex, but I don't think it should be part of their main conclusions.

We appreciate that TWCO's general principles are frequency-agnostic and can be viewed as binding by synchrony in a broad sense. Our work, however, specifically instantiates these principles in V1 gamma: the model reflects TWCO dynamics together with V1 anatomy/physiology and the well-established contrast-frequency relationship in the gamma range (which, to our knowledge, has not been demonstrated with comparable specificity for other bands). In that sense, it is a gamma oscillator model of V1, rather than a generic BBS instantiation. Moreover, stimulus dependencies often cited as challenges to BBS have been used in particular to argue against gamma; showing that these very dependencies are integral to the TWCO mechanism is central to our contribution, and we therefore keep our conclusions focused on the gamma-specific instantiation tested here.

(2) Mediation of the observed effects by the visual features of the figure:

The authors motivate the hypothesis that BBS predicts that the perception of texture-defined objects depends on the density of texture elements and their contrast

heterogeneity. This hypothesis seems trivial as those are the features that distinguish figure from ground. I think it would be important to clarify how this hypothesis is unique to BBS and not explained by competing theories, such as Binding by Enhanced Firing (Roelfsema, 2023). The authors should be clear about what part of the hypothesis is not trivial based on the task and clearly attributable to oscillators and synchrony.

Our stimulus features were derived from theory rather than psychophysical literature. Starting from the principles of TWCO, we mapped frequency detuning and coupling strength onto known anatomical and physiological properties of early visual cortex, and only then derived the corresponding stimulus manipulations (contrast heterogeneity and grid coarseness). We agree that grid coarseness (element distance) is an established facilitator of figure-ground perception. By contrast, contrast heterogeneity (feature variance) is less commonly emphasized as a figure-ground cue, compared to mean-based cues, but follows directly from TWCO's frequency detuning. Importantly, mean contrast and luminance were matched exactly between figure and background in our stimuli. Demonstrating that contrast heterogeneity and grid coarseness not only independently affect figure-ground perception, but reflect a trade-off where higher heterogeneity needs to be counteracted by reduced grid coarseness in the way TWCO specifies is therefore non-obvious and provides an initial empirical indication that the core TWCO variables might shape perception. We also agree that alternative models would further clarify the unique explanatory value of synchrony. In the revised manuscript, we compare rate-based readouts (mean figure rate; figure-background rate difference) with the synchrony readout from the same simulations. Rate difference indeed constitutes a predictor of performance, but the synchrony readout showed a larger effect and was preferred by out-of-sample model comparison.

Using a linear model, the authors assess the relationship between discrimination accuracy and synchrony. Did the authors also include the factors grid coarseness and contrast heterogeneity in this model? Again, as both the task performance (as shown by the GEE analysis) and oscillatory synchrony depend on these features, the relationship between model and behavioral performance will be mediated by the visual features.

Thank you for raising this. In our framework, detuning (via contrast heterogeneity) and coupling (via grid coarseness) are the inputs, synchrony is the proposed mechanistic mediator, and behavior is the output. Because synchrony in our model is a (near-)deterministic function of the manipulated features under fixed parameters, a joint features+synchrony regression is statistically ill-posed (perfect multicollinearity up to numerical error) and cannot add information. A proper mediation test would require trial-wise neural measurements of synchrony in the same task, which we do not have and acknowledge as a limitation in the Discussion. Accordingly, we show that both the features themselves (reflecting TWCO principles) and model-derived synchrony (realizing the proposed pathway) account for behavior.

We agree this does not establish a unique contribution of synchrony. To probe alternatives, we added rate-based readouts and a model comparison to the revised manuscript. These additional analyses indicate that synchrony outperforms simple rate-based mappings. We do not claim this rules out more sophisticated rate-based mechanisms. Our aim is to demonstrate that synchrony is a viable, behaviorally informative readout for downstream processing. We do not assert it is the only mechanism the brain uses. Synchrony had been discounted due to its stimulus dependence; our results are intended to rule it back in. We have made changes throughout the manuscript to better reflect this more modest aim.

(3) Goodness of fit measures are not established a priori:

I have described this concern in my public review. It is hard to assess what the authors would have interpreted as a good or a bad fit, especially without accounting for the confound in the relationship between oscillator synchrony and behavior. Similarly, when

assessing the similarity between the behavioral and dynamic Arnold Tongues across different coupling parameters, the authors found that the chosen parameters (based on macaque data) were not optimal. They offer the explanation that the human cortex has a lower coupling decay than the macaque cortex, and the similarity is higher for lower values of coupling decay. While this explanation is not entirely implausible, it is unclear where an oscillator model with human values would be in the presented plot, as the authors didn't estimate those values from the human studies. Moreover, the task used in the Lowet et al., 2017 paper is very different from the task presented here, which could also account for differences. Overall, the explanation appears hand-wavy considering the lack of empirically defined goodness of fit measures.

Thank you for these concerns.

We did indeed not provide a priori thresholds for what would be considered good fit. Instead, we used two complementary benchmarks; namely noise ceilings and parameter exploration. The former provides an upper bound on what any model (not just ours but based on completely different mechanisms) could achieve given our data. The parameter sweep provides an indication how well our concrete model can maximally fit the data and how bad it can be based on possible parameters. These benchmarks are more informative than a fixed a-priori cutoff, which would depend on unknown noise and inter-subject variability. Both the noise ceiling and the parameter exploration indicate that our model, using a priori fixed parameters, performs well. Additionally, we redid all our statistical analyses after z-normalizing every predictor to provide easier interpretation of effect sizes.

Regarding the reason that key model parameters were not optimal, we believe our interpretation to be plausible. We agree that we currently do not have data to estimate the exact human decay factor and hence cannot establish how much model fit would be affected. However, the parameter exploration in Figure 3 shows that small to modest reductions in decay would improve model fit. We discuss this now in the revised manuscript.

The reviewer's suggestion is intriguing. While Lowet et al. (2017) used a different task, the parameters we took from their work (decay rate and maximum coupling) are intended to reflect anatomical properties and thus should not be task-dependent. That said, Lowet et al.'s data carry uncertainty, so our estimates may not be exact; we note this explicitly in the revised Discussion. Whether a different task would have yielded better parameter estimates is difficult to determine, but we considered Lowet's paradigm appropriate because it was designed to target the same V1 anatomical and physiological properties that map onto TWCO.

I have concerns about a similar confound in the training effects. If I'm not mistaken, the Hebbian Learning rule encourages synchronization between the oscillators in the grid. As such, it causes synchronization to increase over several simulations. Clearly, the task performance of the participants also improves over the sessions. Again, an empirical threshold would be required to assess whether the similarity in learning between model and performance goes beyond what is expected based on learning alone. How much of these effects can be attributed to the model being oscillatory?

The reviewer is correct that, in our framework, learning operates via changes in coupling that increase synchrony. Enhanced synchrony is the proposed (and in our model also the actual) pathway by which learning impacts behavior. We agree that learning could, in principle, act through pathways other than synchrony. Demonstrating this would not be achieved by a mediation analysis here, because that requires independent, trial-level neural measurements of the candidate pathways (synchrony and alternatives). In the absence of such data, the appropriate approach would be model comparison between competing mechanistic readouts. We have added such a model comparison for a synchrony readout versus two rate-based readouts derived from the same simulations for the first session; i.e., focusing on the pathway from stimulus features to behavior. However, a similar model

comparison is not possible for learning. As we show in the supplementary materials, rate-based readouts of our V1 model are not at all affected by coupling strength. As such, they are insensitive to changes in coupling and are thus not viable as alternative mechanisms to explain performance changes due to learning. A fair test of rate-based alternatives would require building a detailed rate-based figure-ground segregation model that predicts session-wise changes. We agree that this is an important next step but it is also substantial undertaking beyond the scope of the present study.

(4) Similarly, for the comparison of the Arnold Tongue in the transfer session and the early session:

In the first part of the Results section, it says: "Our model rests on the assumption that learning-induced structural changes in early visual cortex are specific to the retinotopic locations of the trained stimuli. We evaluated whether this assumption holds for our human participants using the transfer session following the main training period. [...] If learning is indeed local, participants' performance in the transfer session should resemble that of early training sessions, indicating a reset in performance for the new retinal location."

The authors find that a model fit to session 3 explains the data in the transfer session best and consider this as evidence for the above-stated expectation. Again, it is unclear where the cutoff would have been for a session to be declared as early or late. For instance, had the participants only performed 4 sessions, would the performance be best explained by session 3 or session 1?

A high number of statistical tests are used, which, firstly, need to be corrected for multiple comparisons (did the authors do this?). Secondly, I feel that the regression models could be improved. For instance, the authors fit one model per session and then assess how well each model explains the variance in the transfer session. I think the authors might want to opt for one model with the regressors contrast heterogeneity, grid coarseness, and session (and their interaction). Using this approach, the authors would still be able to assess which session predicts the data best. Similarly, interindividual variability could be accounted for by adding participant-specific random effects to the model (and using a mixed model), instead of fitting individual models per participant.

We agree the “early vs late” cutoff was underspecified. In the revision, we predefine Session 2 as the early-learning reference, excluding Session 1 to avoid familiarization/response-mapping effects. We then fit a single Bayesian hierarchical model with contrast heterogeneity, grid coarseness, and session, plus a transfer indicator, and participant-level random effects. This allows us to place the transfer session on the same scale as training and to test a) whether the transfer session precedes the state in session 2 via the posterior contrast $P(\beta_{\text{transfer}} < \beta_{\text{Sess2}})$ and b) whether it is indistinguishable from the state in session two using an equivalence test derived from the fitted model. We find that the transfer session is equivalent to session 2. We added this updated analysis of the transfer session in the Results (paragraph 15).

In response to the suggestion to use a hierarchical regression model for analyzing the transfer session, we have decided to use such a model for all our analyses in a Bayesian framework. In this Bayesian framework, inference is based on the joint posterior (credible intervals/equivalence) of all predictors in a model and additional post-hoc multiplicity corrections are not required.

(5) Questions regarding the model:

What does it mean that the grid was "defined in visual space"? How biologically plausible with regard to the retinotopy and organization of the oscillators do the authors claim the model to be?

We are happy to clarify this point. We have a total of 400 oscillators reflecting neural assemblies in V1. We start by defining a regular, 20x20, grid of the receptive field (RF) centers of these oscillators inside the figure region. Each oscillator is then also assigned a RF size based on the eccentricity of its RF center. We use the threshold-linear relationship between RF eccentricity and RF size reported in [1] to assign RF sizes. Each oscillator thus has an individual, eccentricity-dependent, RF size.

For the coupling between oscillators, we need to know their cortical distances. We obtain these by first determining the cortical location of each oscillator through a complex-logarithmic topographic mapping of neuronal receptive field coordinates onto the cortical surface [2,3]. For this mapping, we use human parameter values estimated by [4]. From these cortical locations, we then compute pairwise Euclidean distances.

The model thus captures realistic retinotopy, eccentricity-dependent RF sizes, and distance-dependent coupling on the cortical surface. We have adjusted our Methods to make these steps clearer.

(1) Freeman, J., & Simoncelli, E. P. (2011). Metamers of the ventral stream. *Nature neuroscience*, 14(9), 1195-1201.

(2) Balasubramanian, M., & Schwartz, E. L. (2002). The isomap algorithm and topological stability. *Science*, 295(5552), 7. <https://doi.org/10.1126/science.1066234>

(3) Schwartz, E. L. (1980). Computational anatomy and functional architecture of striate cortex: a spatial mapping approach to perceptual coding. *Vision Research*, 20(8), 645–669. <http://www.sciencedirect.com/science/article/pii/0042698980900905>

(4) Polimeni, J. R., Hinds, O. P., Balasubramanian, M., van der Kouwe, A. J. W., Wald, L. L., Dale, A. M., & Schwartz, E. L. (2005). Two-dimensional mathematical structure of the human visuotopic map complex in V1, V2, and V3 measured via fMRI at 3 and 7 Tesla. *Journal of Vision*, 5(8), 898. <https://doi.org/10.1167/5.8.898>

Similarly, do the authors claim that each gabor annuli stimulates a single receptive field in V1?

We hope that with the additional explanation above, it is clearer that there is not a one-to-one mapping. Each oscillator samples the local image by pooling over all Gabor annuli that overlap its receptive field (partially or fully) and computes the average contrast within its RF. Conversely, a single annulus typically overlaps multiple RFs and contributes to each in proportion to the overlap.

I am unsure how the oscillators were organized, if not retinotopically. How is the retinotopic input fed into the non-retinotopically arranged oscillators?

We hope that with the additional explanation above, it is clearer that the network is strictly retinotopic.

The frequency of each oscillator changes according to $\omega=2\pi v$ with $v=25+0.25C$. How were the values for the linear regression in v chosen? Reference?

The slope and intercept parameters for this equation were first reported in [5]. We added the reference to the Methods.

(5) Lowet, E., Roberts, M., Hadjipapas, A., Peter, A., van der Eerden, J., & De Weerd, P. (2015). Input-dependent frequency modulation of cortical gamma oscillations shapes spatial synchronization and enables phase coding. *PLoS computational biology*, 11(2), e1004072.

(6) *Hebbian Learning Rule:*

I am confused about how the effective learning rate $E = \epsilon t$ is calculated. It is said that it is estimated based on the similarity between the second experimental session and the distribution of synchrony after letting the model learn. How can the model learn without knowing epsilon and t?

We agree with the reviewer that our procedure to estimate the effective learning rate requires further clarification. We performed a nested grid search. Essentially, we let the model learn between session 1 and 2 with each of 25 candidate effective learning rates and evaluate how well each of them allow the model to fit performance in session 2. We then select the best effective learning rate and create a new, smaller, grid around this value and repeat that procedure. In total we perform 5 nested grids to arrive at the final effective learning rate. We expanded the explanation in the Methods.

(B) *Minor concerns:*

(1) Small N: 2/3 of the studies that were cited to justify the small sample were notably different from the current experiment, i.e., Intoy 2020 is an eye movement task, Lange 2020 is a memory task (Tesileanu 2020 is more similar). I think a power analysis would be great to support, as the sample size seems quite low

Our study uses a within-subject design with ~750 trials per session (~6,000 total) per participant, analyzed with a hierarchical model that pools information across trials and participants. To assess adequacy, we ran a simulation-based design analysis using the fitted hierarchical model (i.e., post hoc, based on the observed variance components). This analysis indicated a detection probability >90% for all key effects. We now report the results of this design analysis in the (Supplementary Table 1) and note this in the Results (paragraph 1).

Regarding the literature context, we agree the cited studies are not identical to ours; we referenced them to illustrate a common practice (small N with many trials) when targeting low-level, early-visual mechanisms. Intoy (pattern/contrast sensitivity) and Lange (perceptual learning in early vision) share that focus, while Tesileanu is methodologically closest.

(2) Figure 1 could be more informative and better described in the text. The authors often don't refer to the panels in Figure 1. Maybe it would help to swap a and b to describe the Arnold tongue first? It might also be a good idea to add the coupling strength and frequency detuning axes

We have swapped panels a and b and now refer to each panel in the main text to enhance clarity.

(3) Values of rho (distance - is this degrees visual angle)? Do the authors assume that the size of the stimuli corresponds to receptive fields in V1? If so, how is this justified?

The center-to-center distance between any pair of neighboring annuli is indeed expressed in degrees of visual angle. Rho is a scaling factor for this distance. With rho=1, the center-to-center distance corresponds to the diameter of the annuli; i.e., they touch but do not overlap each other. We do not assume any relation between the size of receptive fields and the size of the annuli. Receptive field sizes in our model are purely determined by their eccentricity and each oscillator can have several annuli within its receptive field while each annulus can fall within several overlapping receptive fields of different oscillators. We believe that the schematic illustration in Figure 1 might have given the impression that each oscillator sees

exactly one annulus and added a note that this is not the case and merely an oversimplification to illustrate the relationship between contrast and intrinsic frequency.

(4) Some equations are embedded in the text, and some are not. It might be easier to find the respective equation if they all have an index. For instance, the authors mention the psychometric function that relates model synchrony and performance in the results section. It would be easier to find if it had an index that the authors could refer to.

We moved this equation as well as the contrast intrinsic frequency mapping from inline to displayed and numbered them.

(5) Is there a reference for "Our model rests on the assumption that learning-induced structural changes in early visual cortex are specific to the retinotopic locations of the trained stimuli"? (If so, it should be cited.)

We added references supporting this assumption.

(6) Figure 2b: colorbar missing label.

We added the label.

Reviewer #2 (Recommendations for the authors):

Cool work!

(1) The reader would benefit from (a single) comprehensive figure that visually explains the entire conceptual framework-from TWCO principles to neural implementation to behavioural predictions-accessible to readers without specialised knowledge of oscillatory dynamics. This will give the paper a greater impact.

We have adjusted Figure 1 in accordance with suggestions made by reviewer 1 and added further explanations to the caption and the Introduction to enhance clarity on how the principles of TWCO relate to neural implementation.

(2) I think this paper would benefit from the audience eLife provides, but the paper could move closer to the audience.

(3) Pride comes before the fall, but I am not the most uninformed reader, and it took me some effort to process everything.

Thank you, we took this to heart. In the Introduction, we now state more explicitly how each variable is operationalized and how these map onto TWCO with improved reference to relevant panels in the schematic figure. We agree the framework is conceptually dense. TWCO principles reach the stimuli through specific V1 anatomy and physiology, so there are several links to keep in mind. Our goal with the revised introduction and figure is to make those links better visible.

(4) You could consider discussing potential implications for understanding perceptual disorders characterized by altered neural synchrony (e.g., schizophrenia, autism) and how your learning paradigm might inform perceptual training interventions.

Thank you for this suggestion. We have added that TWCO might provide a new lens to study perceptual disorders to the Discussion. We provide a concrete example of the relation between grouping, gamma synchrony (in light of TWCO) and lateral connectivity in schizophrenia

(5) I think this paper has real strength, but rather than dispersing limitations throughout the discussion, create a dedicated section that systematically addresses ecological

validity, alternative explanations, and generalisability concerns. This will also preempt criticism.

We appreciate the suggestion. Our preference is to discuss limitations in context, next to the specific results they qualify, so readers see why each limitation matters and how it affects interpretation. Nevertheless, paragraph 7 on page 20 summarizes most limitations in a single paragraph.

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